

CURVATURE OBSTRUCTIONS AND FLAT TOTALLY GEODESIC TORI IN HESSIAN DEFORMATIONS OF PRODUCTS OF SPHERES

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ABSTRACT. Let h be the product of the unit round metrics on $\mathbb{S}^m \times \mathbb{S}^n$, let $A : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{m+1}$ be linear, and put

$$s_A(x, y) = \langle x, Ay \rangle, \quad g_{\gamma, A} = h + \gamma \operatorname{Hess}_h s_A.$$

Writing $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$ for the singular values of A , padded by zeros, we prove that $g_{\gamma, A}$ is Riemannian exactly when $|\gamma|(\sigma_1 + \sigma_2) < 1$. If $m, n \geq 2$ and $A \neq 0$, the top three singular pairs determine a totally geodesic $\mathbb{S}^2 \times \mathbb{S}^2$ containing a plane of negative sectional curvature at every nonzero Riemannian parameter. Thus the round product is the unique nonnegatively curved metric on each such Hessian ray. We also obtain an exact formula for every background-mixed curvature and an explicit dimension-free negative margin on normalized parameter annuli. For every pair of singular directions, the metric contains a flat totally geodesic 2-torus, so no member of the family has strictly positive sectional curvature.

For arbitrary paths $h + t \operatorname{Hess} f + t^2 B + O(t^3)$, we give a two-jet normal form modulo diffeomorphisms. An integrated trace identity forces every quadratic mixed repair inequality to be an equality on a closed product. We classify the resulting coupled kernel modulo gauge and factor variations: one tensor leg must be Killing and the other co-closed. On $\mathbb{S}^2 \times \mathbb{S}^2$, every nonzero coupled class has a nearby plane with curvature $-\delta s^2 + O(s^3)$; this includes cross-degree superpositions, simultaneous mixtures of both one-sided towers, and the full $(1, 1)$ block. The associated canonical corrected Hessian path therefore fails at order t^4 . A transverse rotational sum also admits an exact negative oblique formula. These are obstructions for the stated Hessian family and kernel rays, not a proof of the Hopf conjecture.

1. INTRODUCTION

The question whether $\mathbb{S}^2 \times \mathbb{S}^2$ admits a metric of strictly positive sectional curvature is the classical Hopf conjecture. It is open. A useful way to approach the problem without overstating what a special construction can prove is to choose a geometrically natural family of metrics, determine its full parameter range, and identify an exact obstruction inside that family.

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This paper carries out that program for the finite-dimensional family

$$(1) \quad g_{\gamma,A} = h + \gamma \operatorname{Hess}_h s_A, \quad s_A(x, y) = \langle x, Ay \rangle,$$

on a product of unit spheres. Here $A : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{m+1}$ is arbitrary and $\gamma \in \mathbb{R}$. The functions s_A are precisely the linear combinations of bilinear products of degree-one spherical harmonics. Thus the family includes couplings with no continuous symmetry inherited from the two round factors, but it retains enough structure for the exact global results below.

The results have three parts. First, a sharp positivity criterion depends on the two largest singular values of A . Second, an exact mixed-curvature formula gives a spectral-gap obstruction: when $\sigma_1 > \sigma_2$, a fixed mixed plane has negative curvature at every nonzero point of the Riemannian ray. A top-critical oblique plane in the totally geodesic top-three singular $\mathbb{S}^2 \times \mathbb{S}^2$ is negative for every matrix of rank at least two; a zero-critical oblique plane handles rank one and lies in the same slice. Thus all dimensions are detected by a four-dimensional submodel, with an explicit dimension-free quantitative margin. Third, each pair of singular directions cuts out a product of great circles which remains totally geodesic and becomes intrinsically flat for the deformed metric. Together the two curvature mechanisms show that, whenever $m, n \geq 2$ and $A \neq 0$, the round product is the only metric on its Hessian ray with nonnegative sectional curvature. At the isotropic $\mathbb{S}^2 \times \mathbb{S}^2$ endpoint all background-mixed curvatures are nevertheless nonnegative. This mixed-oblique dichotomy shows in particular that background-mixed nonnegativity is not a proxy for full nonnegative sectional curvature.

The exact rays also admit a useful deformation-theoretic interpretation. Since $\operatorname{Hess} f = \frac{1}{2} \mathcal{L}_{\nabla f} h$, their first variation is pure gauge. We give a two-jet normal form for arbitrary corrected paths $h + t \operatorname{Hess} f + t^2 B + O(t^3)$ and an explicit mixed-curvature functional which every nonnegatively curved path must make nonnegative. Its integrated trace vanishes, so on the closed product all those values must in fact be zero. For $f = s_A$, singular triples turn this into concrete identities for the second-order correction B . The canonical gauge correction satisfies them, whereas the uncorrected ray $B = 0$ violates one for every $A \neq 0$. Modulo diffeomorphisms and factor variations, we classify the full kernel: every coupled class is represented by a pure mixed, divergence-free tensor with a Killing one-form on at least one factor and a co-closed one-form on the other. On $\mathbb{S}^2 \times \mathbb{S}^2$ this leaves exactly the harmonic bidegrees $(1, \ell)$ and $(\ell, 1)$. A two-curl moving-plane calculation and a frame-average spectral gap obstruct every nonzero element of this entire coupled quotient, including arbitrary cross-degree and two-tower mixtures. A signed singular-value argument closes the full $(1, 1)$ block. The canonical transverse rotational sum additionally has an exact negative oblique formula.

Theorem 1.1. *Let $m, n \geq 1$, set $q = \min\{m+1, n+1\}$, and let $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_q \geq 0$ be the singular values of $A : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{m+1}$. If $A \neq 0$, then*

$$g_{\gamma,A} \text{ is Riemannian} \iff |\gamma|(\sigma_1 + \sigma_2) < 1.$$

For every γ in this interval and every choice of singular frames, $g_{\gamma,A}$ contains one flat totally geodesic torus for each pair $1 \leq i < j \leq q$. In particular,

$$\sec_{g_{\gamma,A}}(T_p T_{ij}) = 0 \quad \text{for every } p \in T_{ij},$$

and $g_{\gamma,A}$ never has strictly positive sectional curvature. If moreover $m, n \geq 2$, then the totally geodesic $\mathbb{S}^2 \times \mathbb{S}^2$ determined by the top three singular pairs contains a plane of strictly negative sectional curvature for every nonzero Riemannian parameter. Hence the metric does not have nonnegative sectional curvature. If $\sigma_2 > 0$, the plane can be chosen top-critical oblique; in rank one it can be chosen zero-critical oblique. If $\sigma_1 > \sigma_2$, it can also be chosen background-mixed. If $A = 0$, then $g_{\gamma,A} = h$ for every γ .

The negative mixed curvature in the examples below is strict. It therefore persists under small changes of the matrix, base point, and 2-plane; it is not an isolated algebraic coincidence.

For $\mathbb{S}^2 \times \mathbb{S}^2$, a chosen singular frame gives three such tori. When A is invertible with three distinct singular values, the factor-preserving isometries inherited from the round product form a finite group. This matters because the powerful theorem of Hsiang–Kleiner excludes positive curvature on $\mathbb{S}^2 \times \mathbb{S}^2$ in the presence of an isometric circle action [7]. Our obstruction is instead visible directly on fixed tori of finite involutions. We do not claim that the full isometry group of the deformed metric is always finite.

Deformations of product metrics have long been studied in connection with Hopf’s question, notably by Bourguignon [5, 4]. More recent work has produced obstructions under symmetry assumptions [7, 2] and within conformal or warped classes [8]; see also the survey [1]. The present result is deliberately narrower and fully explicit: it gives a complete positive-curvature obstruction for the family (1), not the Hopf conjecture. Its additional contribution is that the same mechanism works on every product $\mathbb{S}^m \times \mathbb{S}^n$ and identifies the exact Riemannian interval rather than only a perturbative neighborhood of the product metric.

The proof is organized as follows. Section 2 gives an intrinsic formula for the Hessian and its tangent-space block form. Section 3 proves the sharp positivity interval, derives the exact mixed-curvature formula, proves the spectral-gap and top-critical oblique obstructions, isolates the isotropic mixed–oblique dichotomy, and analyzes plane-wise sign-change thresholds. Section 4 constructs and analyzes the flat totally geodesic tori. Section 5 separates the inherited factor-preserving symmetry from the generally harder full isometry group. Section 6 gives the two-jet normal form and the integrated rigidity, explicit repair identities, the full coupled kernel classification, the nonlinear obstruction for its entire coupled quotient, and an exact transverse rotational model. Section 7 records the precise relation to the Hopf problem and to possible extensions.

2. THE HESSIAN DEFORMATION

Let

$$M = \mathbb{S}^m \times \mathbb{S}^n \subset \mathbb{R}^{m+1} \times \mathbb{R}^{n+1}, \quad h = h_m \oplus h_n,$$

where both factors have radius one. At $(x, y) \in M$ we identify

$$T_{(x,y)}M = x^\perp \oplus y^\perp.$$

The Levi-Civita connection and Hessian below are those of h .

Proposition 2.1 (Hessian formula). *For $X, X' \in x^\perp$ and $Y, Y' \in y^\perp$,*

$$(2) \quad \begin{aligned} \text{Hess } s_A((X, Y), (X', Y')) &= -s_A(x, y)(\langle X, X' \rangle + \langle Y, Y' \rangle) \\ &\quad + \langle X, AY' \rangle + \langle X', AY \rangle. \end{aligned}$$

Equivalently, with P_x and P_y the orthogonal projections onto $x^\perp$ and $y^\perp$, respectively,

$$(3) \quad g_{\gamma, A} = \begin{pmatrix} \lambda I_{x^\perp} & \gamma P_x A P_y \\ \gamma P_y A^\top P_x & \lambda I_{y^\perp} \end{pmatrix}, \quad \lambda = 1 - \gamma s_A(x, y).$$

Proof. For a fixed vector $a \in \mathbb{R}^{m+1}$, the restriction of the linear function $x \mapsto \langle x, a \rangle$ to the unit sphere satisfies

$$(4) \quad \text{Hess}_{\mathbb{S}^m} \langle x, a \rangle = -\langle x, a \rangle h_m.$$

Indeed, the tangential gradient is $a - \langle x, a \rangle x$, and differentiating it with the Gauss formula gives (4). Holding y fixed and taking $a = Ay$ yields the first pure block of (2); holding x fixed gives the second. Since the product connection differentiates independently in the two factors, the mixed derivative is

$$X(\langle x, AY' \rangle) = \langle X, AY' \rangle.$$

Symmetry of the Hessian supplies the other mixed entry. Adding h gives (3). $\square$

Formula (3) already exhibits the two quantities that compete for definiteness: the scalar diagonal coefficient $1 - \gamma s_A$ and the largest singular value of the tangent compression $P_x A P_y$. The next section optimizes their combination globally.

3. THE SHARP RIEMANNIAN RANGE AND MIXED CURVATURE

We use a two-frame form of the Ky Fan variational principle. Singular values are always listed in nonincreasing order and padded by zeros.

Lemma 3.1 (Two-frame principle). *Let $A : \mathbb{R}^r \rightarrow \mathbb{R}^p$ have singular values $\sigma_1 \geq \sigma_2 \geq \dots$. Then*

$$(5) \quad \sup_{\substack{x, X \in \mathbb{R}^p \text{ orthonormal} \\ y, Y \in \mathbb{R}^r \text{ orthonormal}}} (\langle x, Ay \rangle + |\langle X, AY \rangle|) = \sigma_1 + \sigma_2.$$

The same supremum is obtained with $-\langle x, Ay \rangle$ in place of $\langle x, Ay \rangle$.

Proof. Change the sign of X if necessary so that the second pairing is nonnegative. The expression is then the trace pairing of A with a rank-two partial isometry. The Ky Fan trace inequality bounds it by $\sigma_1 + \sigma_2$; equivalently, this is the rank-two case of the variational characterization of the nuclear norm of a compression [6, Chapter III]. Equality is obtained from left and right singular vectors $x = u_1$, $X = u_2$, $y = v_1$, and $Y = v_2$. Replacing x by $-x$ proves the final assertion. $\square$

For completeness, define the largest tangent coupling at (x, y) by

$$(6) \quad \tau(x, y) = \max_{\substack{X \perp x, Y \perp y \\ \|X\|=\|Y\|=1}} |\langle X, AY \rangle|.$$

Lemma 3.1 says precisely that

$$(7) \quad \sup_{x,y} (s_A(x, y) + \tau(x, y)) = \sup_{x,y} (\tau(x, y) - s_A(x, y)) = \sigma_1 + \sigma_2.$$

Proposition 3.2 (Exact definiteness interval). *If $A \neq 0$, then $g_{\gamma,A}$ is positive definite at every point of M if and only if*

$$(8) \quad |\gamma|(\sigma_1 + \sigma_2) < 1.$$

At either endpoint the tensor is degenerate somewhere, and beyond an endpoint it is not positive semidefinite everywhere.

Proof. At a fixed (x, y) , choose singular frames for the tangent compression $P_x A P_y$. The block matrix (3) is positive definite if and only if

$$(9) \quad \lambda > |\gamma|\tau(x, y), \quad \lambda = 1 - \gamma s_A(x, y).$$

The strict inequality also implies $\lambda > 0$.

Suppose first that $\gamma > 0$. By (7),

$$\lambda - \gamma\tau = 1 - \gamma(s_A + \tau) \geq 1 - \gamma(\sigma_1 + \sigma_2).$$

For $\gamma < 0$, writing $a = |\gamma|$ gives

$$\lambda - a\tau = 1 - a(\tau - s_A) \geq 1 - a(\sigma_1 + \sigma_2).$$

Thus (8) implies positive definiteness.

Conversely, take singular vectors satisfying $Av_k = \sigma_k u_k$. For $\gamma > 0$, at $(x, y) = (u_1, v_1)$ evaluate the metric on $(u_2, -v_2)$. Proposition 2.1 gives

$$(10) \quad g_{\gamma,A}((u_2, -v_2), (u_2, -v_2)) = 2(1 - \gamma(\sigma_1 + \sigma_2)).$$

For $\gamma = -a < 0$, use $(x, y) = (-u_1, v_1)$ and the vector (u_2, v_2) to obtain

$$(11) \quad g_{-a,A}((u_2, v_2), (u_2, v_2)) = 2(1 - a(\sigma_1 + \sigma_2)).$$

These expressions vanish at the endpoints and are negative beyond them. Padding the singular values by zero makes the argument include rank-one maps without modification. $\square$

Remark 3.3. The occurrence of $\sigma_1 + \sigma_2$, rather than only the operator norm σ_1 , is geometric. One top singular direction fixes the base point (x, y) and contributes through s_A ; a second orthogonal direction remains available in the tangent spaces and contributes through the mixed Hessian block.

We next calculate every plane spanned by a vector from each background factor. The formula also explains why a nonnegative second-order term does not imply nonnegative curvature for small nonzero parameters.

Theorem 3.4 (Exact background-mixed curvature). *Fix $p = (x, y)$ and put*

$$a = P_x A y \in x^\perp, \quad b = P_y A^\top x \in y^\perp.$$

For $X \in x^\perp$ and $Y \in y^\perp$, set

$$r = \|X\|^2, \quad w = \|Y\|^2, \quad \alpha = \langle a, X \rangle, \quad \beta = \langle b, Y \rangle,$$

and, using h to identify vectors and covectors, define

$$(12) \quad \begin{aligned} \eta_{XX} &= \left(\frac{r}{2}a - \alpha X, -\frac{r}{2}b\right), \\ \eta_{XY} &= \left(-\frac{\beta}{2}X, -\frac{\alpha}{2}Y\right), \\ \eta_{YY} &= \left(-\frac{w}{2}a, \frac{w}{2}b - \beta Y\right). \end{aligned}$$

Whenever $g_{\gamma,A}$ is Riemannian,

$$(13) \quad \sec_{g_{\gamma,A}}(X \wedge Y) = \frac{\gamma^2}{\Delta_{\gamma,A}(X, Y)} \left(g_{\gamma,A}^{-1}(\eta_{XY}, \eta_{XY}) - g_{\gamma,A}^{-1}(\eta_{XX}, \eta_{YY}) \right),$$

where

$$\Delta_{\gamma,A}(X, Y) = g_{\gamma,A}(X, X)g_{\gamma,A}(Y, Y) - g_{\gamma,A}(X, Y)^2.$$

In particular, if X and Y are h -unit vectors, then

$$(14) \quad \sec_{g_{\gamma,A}}(X \wedge Y) = \frac{\gamma^2}{4} \left(\|a\|^2 - \alpha^2 + \|b\|^2 - \beta^2 \right) + O(\gamma^3).$$

Proof. Write $H = \text{Hess}_h s_A$ and choose h -normal coordinates at p adapted to $x^\perp \oplus y^\perp$. If Γ_{WUV}^g denotes the Christoffel symbol of the first kind, then at p

$$\begin{aligned} \Gamma_{WUV}^g &= \gamma \mathcal{C}(U, V, W), \\ \mathcal{C}(U, V, W) &= \frac{1}{2} ((\nabla_U H)(V, W) + (\nabla_V H)(U, W) \\ &\quad - (\nabla_W H)(U, V)). \end{aligned}$$

In the lower-index coordinate formula for curvature, the part involving second derivatives of the metric is affine in γ . Its constant term vanishes on a background-mixed plane because the round product has zero mixed curvature. Its linear term vanishes as well: indeed, $H = \frac{1}{2} \mathcal{L}_{\nabla s_A} h$, so the linearized curvature is a Lie derivative of the product curvature tensor. In each term evaluated on (X, Y, X, Y) at least one vector remains in each factor, and the product curvature tensor vanishes. Consequently the exact lower-index curvature is only the Christoffel-quadratic part,

$$(15) \quad R^g(X, Y, X, Y) = \gamma^2 \left(g^{-1}(\mathcal{C}_{XY}, \mathcal{C}_{XY}) - g^{-1}(\mathcal{C}_{XX}, \mathcal{C}_{YY}) \right).$$

It remains to calculate the three covectors. Differentiating Proposition 2.1 with $U = (U_1, U_2)$ gives

$$\begin{aligned} (\nabla_U H)(V, W) = & -ds_A(U)h(V, W) - \langle U_1, V_1 \rangle \langle b, W_2 \rangle - \langle U_2, W_2 \rangle \langle a, V_1 \rangle \\ & - \langle U_1, W_1 \rangle \langle b, V_2 \rangle - \langle U_2, V_2 \rangle \langle a, W_1 \rangle. \end{aligned}$$

Substitution of (X, X) , (X, Y) , and (Y, Y) yields exactly the covectors in (12). Dividing (15) by the squared g -area proves (13). Setting $g^{-1} = h^{-1} + O(\gamma)$ and using $r = w = 1$ gives (14). $\square$

The nonnegative coefficient in (14) does not control the sign on a punctured neighborhood of the product metric. In fact, a simple largest singular value forces strict negative curvature along the entire punctured Riemannian ray.

Theorem 3.5 (Generic spectral-gap obstruction). *Assume $m, n \geq 2$ and $\sigma_1 > \sigma_2$. Write*

$$s = \sigma_1, \quad t = \sigma_2, \quad \rho = \sigma_3,$$

where singular values are padded by zeros, and set

$$(16) \quad c^2 = \frac{s^2 + t^2 - 2\rho^2}{2(s^2 - \rho^2)}, \quad d^2 = \frac{s^2 - t^2}{2(s^2 - \rho^2)}.$$

Choose positive square roots and singular pairs satisfying $Av_i = \sigma_i u_i$ and $A^\top u_i = \sigma_i v_i$, and define

$$(17) \quad \begin{aligned} x &= cu_1 + du_3, & y &= v_2, & X &= (u_2, 0), \\ B^2 &= \frac{s^2 + t^2}{2}, & Y &= \frac{scv_1 + \rho dv_3}{B}, & Y &= (0, Y). \end{aligned}$$

Then X, Y are h -orthonormal at $p = (x, y)$ and, for every γ in the Riemannian interval,

$$(18) \quad \sec_{g_{\gamma, A}}(X \wedge Y) = -\frac{\gamma^4(s^2 - t^2)^2(s^2 + t^2 - 2\rho^2)}{16(s^2 + t^2)\left(1 - \frac{\gamma^2}{2}(s^2 - t^2 + 2\rho^2)\right)}.$$

In particular, this curvature is strictly negative for every $\gamma \neq 0$ in that interval.

Proof. The numbers in (16) are positive and add to one. At the point and plane in (17), the quantities in Theorem 3.4 are

$$a = tu_2, \quad b = BY, \quad \alpha = t, \quad \beta = B.$$

Put

$$Z = du_1 - cu_3, \quad k = sdv_1 - \rho cv_3,$$

and abbreviate

$$(19) \quad \mu = \langle k, Y \rangle = \frac{cd(s^2 - \rho^2)}{B}, \quad \kappa^2 = \|k\|^2 = \frac{s^2 - t^2 + 2\rho^2}{2}.$$

Since $s_A(x, y) = 0$, the diagonal coefficient of the metric at p is one. The tangent compression $C = P_x A P_y$ satisfies

$$C^\top u_2 = 0, \quad Cw = \langle k, w \rangle Z \quad \text{for } w \in \text{span}\{v_1, v_3\}.$$

The remaining singular directions form orthogonal blocks and do not couple to u_2 or Y . The Schur-complement formula, or a direct inversion of the block spanned by Z and k , therefore gives

$$(20) \quad \begin{aligned} g_{\gamma,A}^{-1}((u_2, 0), (u_2, 0)) &= 1, \\ g_{\gamma,A}^{-1}((0, Y), (0, Y)) &= 1 + \frac{\gamma^2 \mu^2}{1 - \gamma^2 \kappa^2} =: q_\gamma, \\ g_{\gamma,A}^{-1}((u_2, 0), (0, Y)) &= 0. \end{aligned}$$

The denominator in (20) is positive throughout the Riemannian interval; this also follows directly from positive definiteness of the corresponding principal block.

For this plane the three covectors in (12) reduce to

$$\eta_{XX} = \eta_{YY} = \left(-\frac{t}{2}u_2, -\frac{B}{2}Y \right), \quad \eta_{XY} = \left(-\frac{B}{2}u_2, -\frac{t}{2}Y \right).$$

Moreover, the squared $g_{\gamma,A}$ -area of the plane is one. Hence Theorem 3.4 yields

$$\begin{aligned} \sec_{g_{\gamma,A}}(X \wedge Y) &= \frac{\gamma^2}{4} (B^2 + t^2 q_\gamma - t^2 - B^2 q_\gamma) \\ &= -\frac{\gamma^4 (B^2 - t^2) \mu^2}{4(1 - \gamma^2 \kappa^2)}. \end{aligned}$$

Finally,

$$B^2 - t^2 = \frac{s^2 - t^2}{2}, \quad \mu^2 = \frac{(s^2 + t^2 - 2\rho^2)(s^2 - t^2)}{2(s^2 + t^2)}.$$

Substitution gives (18). Every factor in its numerator is positive when $s > t \geq \rho$, which proves the sign claim. $\square$

Corollary 3.6 (Generic isolation from nonnegative curvature). *For $m, n \geq 2$, the matrices with $\sigma_1 > \sigma_2$ form an open dense subset of $\text{Hom}(\mathbb{R}^{n+1}, \mathbb{R}^{m+1})$. For every such A ,*

$$\left\{ \gamma : g_{\gamma,A} \text{ is Riemannian and } \sec_{g_{\gamma,A}} \geq 0 \right\} = \{0\}.$$

Thus the round product is isolated on every generic bilinear Hessian ray inside the cone of nonnegatively curved metrics.

Proof. Having a simple largest singular value is an open dense condition on matrices. At $\gamma = 0$ the metric is the nonnegatively curved round product. For every other Riemannian parameter, Theorem 3.5 supplies a plane of strictly negative sectional curvature. $\square$

The spectral-gap theorem therefore leaves the repeated-top stratum as the only possible source of a nontrivial nonnegatively curved metric. At its maximally symmetric endpoint on $\mathbb{S}^2 \times \mathbb{S}^2$, the mixed curvature sign reverses completely: no background-mixed plane is negative.

Theorem 3.7 (Isotropic mixed-curvature nonnegativity). *Let $m = n = 2$ and $A = \sigma Q$, where $\sigma > 0$ and $Q \in O(3)$. Put $\delta = \gamma\sigma$, so the Riemannian interval is $|\delta| < 1/2$. Fix a point (x, y) and background-unit tangent vectors $X \perp x$ and $Y \perp y$. Complete them to orthonormal frames*

$$\mathcal{E} = (x, X, \hat{X}), \quad \mathcal{F} = (y, Y, \hat{Y})$$

whose orientations are chosen so that $R = \mathcal{E}^\top Q \mathcal{F} \in SO(3)$. Represent R by a unit quaternion

$$q_0 + q_1 \mathbf{i} + q_2 \mathbf{j} + q_3 \mathbf{k}$$

and set

$$\omega = q_0^2, \quad \xi = q_1^2, \quad v = q_2^2, \quad \zeta = q_3^2.$$

Then $\omega + \xi + v + \zeta = 1$ and

$$(21) \quad \sec_{g_{\gamma,A}}(X \wedge Y) = \frac{2\delta^2(1 - 2\delta R_{00})}{\det(G_{\delta,R}) \Delta_{\delta,R}(X, Y)} \mathcal{S}_\delta,$$

where $G_{\delta,R}$ is the matrix of $g_{\gamma,A}$ in the adapted tangent frame, $\Delta_{\delta,R}(X, Y)$ is the squared area of the plane, and

$$(22) \quad \begin{aligned} \mathcal{S}_\delta &= \xi\zeta(1 - 2\delta\xi)(1 + 2\delta\zeta) + \omega v(1 - 2\delta\omega)(1 + 2\delta v) \\ &\quad + 8\delta^2\omega\xi v\zeta. \end{aligned}$$

Consequently every background-mixed sectional curvature is nonnegative throughout the Riemannian interval. If $\delta \neq 0$, equality holds precisely when

$$\xi\zeta = 0, \quad \omega v = 0.$$

Proof. Scaling A replaces γ by δ , so it is enough to work with Q . Relative to the adapted frames, write

$$R = \begin{pmatrix} s & \beta & b_\perp \\ \alpha & p & q \\ a_\perp & r & z \end{pmatrix}, \quad C = \begin{pmatrix} p & q \\ r & z \end{pmatrix}, \quad \lambda = 1 - \delta s.$$

The tangent metric has matrix

$$(23) \quad G_{\delta,R} = \begin{pmatrix} \lambda I_2 & \delta C \\ \delta C^\top & \lambda I_2 \end{pmatrix}.$$

The three covectors in Theorem 3.4 are

$$\begin{aligned} \eta_{XX} &= \tfrac{1}{2}(-\alpha, a_\perp, -\beta, -b_\perp), & \eta_{XY} &= \tfrac{1}{2}(-\beta, 0, -\alpha, 0), \\ \eta_{YY} &= \tfrac{1}{2}(-\alpha, -a_\perp, -\beta, b_\perp). \end{aligned}$$

A direct block inversion gives

$$(24) \quad G_{\delta,R}^{-1}(\eta_{XY}, \eta_{XY}) - G_{\delta,R}^{-1}(\eta_{XX}, \eta_{YY}) = \frac{N}{4D},$$

where $D = \det(G_{\delta,R}) > 0$ and

$$\begin{aligned} N = & a_{\perp}^2 \lambda(\lambda^2 - \delta^2(p^2 + q^2)) + b_{\perp}^2 \lambda(\lambda^2 - \delta^2(p^2 + r^2)) \\ & + 2a_{\perp} b_{\perp} \delta(\lambda^2 z - \delta^2 p(pz - qr)) \\ & + \lambda \delta^2 (\beta^2 - \alpha^2)(q^2 - r^2). \end{aligned}$$

Use the standard quaternion matrix

$$R = \begin{pmatrix} 1 - 2(q_2^2 + q_3^2) & 2(q_1 q_2 - q_0 q_3) & 2(q_1 q_3 + q_0 q_2) \\ 2(q_1 q_2 + q_0 q_3) & 1 - 2(q_1^2 + q_3^2) & 2(q_2 q_3 - q_0 q_1) \\ 2(q_1 q_3 - q_0 q_2) & 2(q_2 q_3 + q_0 q_1) & 1 - 2(q_1^2 + q_2^2) \end{pmatrix}.$$

Substitution into the preceding numerator, followed only by the unit relation $q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1$, gives

$$(25) \quad N = 8(1 - 2\delta R_{00})\mathcal{S}_{\delta}.$$

Equations (13), (24), and (25) prove (21).

It remains only to read the sign. Since $|\delta| < 1/2$ and each of ω, ξ, v, ζ lies in $[0, 1]$, every linear factor in (22) is positive. Also $1 - 2\delta R_{00} > 0$, while D and the squared area are positive by the Riemannian assumption. Thus the curvature is nonnegative. For $\delta \neq 0$, the sum vanishes exactly when its first two products vanish, which is the stated equality condition. $\square$

The oblique obstruction propagates to arbitrary dimensions by a fixed-point reduction that will also underlie the flat-torus construction.

Proposition 3.8 (Singular-subsphere reduction). *Choose a singular frame*

$$Av_i = \sigma_i u_i, \quad A^T u_i = \sigma_i v_i, \quad 1 \leq i \leq q,$$

and let $I \subset \{1, \dots, q\}$ have cardinality $r \geq 2$. Set

$$N_I = (\mathbb{S}^m \cap \text{span}\{u_i : i \in I\}) \times (\mathbb{S}^n \cap \text{span}\{v_i : i \in I\}).$$

Whenever $g_{\gamma,A}$ is Riemannian, $N_I \cong \mathbb{S}^{r-1} \times \mathbb{S}^{r-1}$ is totally geodesic. Its induced metric is precisely the Hessian family associated to the diagonal restriction

$$A_I : \text{span}\{v_i : i \in I\} \longrightarrow \text{span}\{u_i : i \in I\}, \quad A_I v_i = \sigma_i u_i.$$

Consequently every sectional-curvature obstruction in a singular submodel is also an obstruction in the ambient product.

Proof. Let R_I be $+I$ on $\text{span}\{u_i : i \in I\}$ and $-I$ on its orthogonal complement, and define S_I analogously using the v_i . The singular decomposition gives

$$R_I^T A S_I = A.$$

Thus $(x, y) \mapsto (R_I x, S_I y)$ preserves h and s_A , hence also $g_{\gamma,A}$. Its fixed-point set is N_I , so the fixed-point lemma for isometries makes N_I totally geodesic. The product of great subspheres is also totally geodesic for h . Therefore the restriction of $\text{Hess}_h s_A$ is the intrinsic Hessian of $s_A|_{N_I}$, which is the

bilinear potential determined by A_I . The final assertion follows from the Gauss equation. $\square$

The isotropic theorem concerns planes mixed with respect to the background product splitting. The next calculation gives a negative oblique plane at the top singular critical point for every matrix of rank at least two, not only on the repeated-top stratum.

Proposition 3.9 (Top-critical oblique-plane obstruction). *Let $m, n \geq 2$, let $\gamma > 0$, and abbreviate*

$$x = \gamma\sigma_1, \quad y = \gamma\sigma_2, \quad z = \gamma\sigma_3, \quad d = 1 - x.$$

Choose ordered singular pairs $Av_i = \sigma_i u_i$ and set

$$\lambda_i^\pm = d \pm \gamma\sigma_i, \quad \Delta_i = \lambda_i^+ \lambda_i^-, \quad i = 2, 3.$$

Throughout the Riemannian range $x + y < 1$, the four vectors at (u_1, v_1)

$$E_i^+ = \frac{(u_i, v_i)}{\sqrt{2\lambda_i^+}}, \quad E_i^- = \frac{(u_i, -v_i)}{\sqrt{2\lambda_i^-}}, \quad i = 2, 3,$$

are $g_{\gamma,A}$ -orthonormal. The plane spanned by

$$(26) \quad U = \frac{E_2^+ - E_2^-}{\sqrt{2}}, \quad V = \frac{E_3^+ + E_3^-}{\sqrt{2}}$$

has unit squared area and

$$(27) \quad \sec_{g_{\gamma,A}}(U \wedge V) = -\frac{\mathcal{N}(x, y, z)}{2\Delta_2\Delta_3},$$

where

$$(28) \quad \mathcal{N}(x, y, z) = d(y^2 + z^2) + (2x - 1)(d^2 - \sqrt{\Delta_2\Delta_3}).$$

If $\sigma_2 > 0$, this curvature is strictly negative. The same conclusion holds for $\gamma < 0$ after applying the antipodal map on the first sphere.

On the repeated-top stratum $x = y = \delta$ and $z = \tau\delta$, put

$$a = 1 - 2\delta, \quad \Pi = (1 - \delta)^2 - \tau^2\delta^2.$$

Then (27) becomes

$$(29) \quad \sec_{g_{\gamma,A}}(U \wedge V) = \frac{(1 - \delta)(\Pi - \delta) - a\sqrt{a\Pi}}{2a\Pi}.$$

In particular, at the isotropic endpoint $\tau = 1$ it is $-\delta^2/(2(1 - 2\delta)^2)$.

Proof. Proposition 3.8 reduces the computation to the totally geodesic top-three singular $\mathbb{S}^2 \times \mathbb{S}^2$. At (u_1, v_1) the tangent metric splits into symmetric and antisymmetric modes with coefficients λ_i^+ and λ_i^- , proving orthonormality. Direct differentiation in orthographic coordinates gives unit area and (27)–(28).

It remains to read the sign. Put

$$S = y^2 + z^2, \quad L = \sqrt{(d^2 - y^2)(d^2 - z^2)}.$$

The Riemannian inequality gives $d > y \geq z \geq 0$. Hence $0 < L \leq d^2$. If $x \geq 1/2$, then $\mathcal{N} \geq dS > 0$. If $x < 1/2$, the elementary inequality

$$1 - \sqrt{(1-r)(1-s)} \leq r + s, \quad 0 \leq r, s < 1,$$

applied to $r = y^2/d^2$ and $s = z^2/d^2$ gives $d^2 - L \leq S$. Consequently

$$\mathcal{N} = dS - (1 - 2x)(d^2 - L) \geq xS > 0.$$

Since $S > 0$ exactly when $y > 0$, the strict sign follows. The map $(x_0, y_0) \mapsto (-x_0, y_0)$ pulls $g_{\gamma, A}$ back to $g_{-\gamma, A}$, proving the negative-parameter assertion. Substitution of $x = y = \delta$ and $z = \tau\delta$ gives the final two formulas. $\square$

Proposition 3.10 (Rank-one zero-critical obstruction). *Assume $m, n \geq 2$ and A has rank one. Write its only positive singular value as σ , choose singular pairs so that $\sigma_1 = \sigma$ and $\sigma_2 = \sigma_3 = 0$, and put $\delta = \gamma\sigma$. At the zero critical point $p = (u_2, v_2)$ define*

$$F_1^\pm = \frac{(u_1, \pm v_1)}{\sqrt{2(1 \pm \delta)}}, \quad F_3^\pm = \frac{(u_3, \pm v_3)}{\sqrt{2}}.$$

For $|\delta| < 1$, the vectors

$$U = \frac{F_1^+ - F_1^-}{\sqrt{2}}, \quad V = \frac{F_3^+ + F_3^-}{\sqrt{2}} = (u_3, 0)$$

are $g_{\gamma, A}$ -orthonormal and

$$(30) \quad \sec_{g_{\gamma, A}}(U \wedge V) = -\frac{1 - \sqrt{1 - \delta^2}}{2\sqrt{1 - \delta^2}}.$$

Thus this plane is strictly negative for every nonzero Riemannian parameter, and

$$(31) \quad \sec_{g_{\gamma, A}}(U \wedge V) = -\frac{\gamma^2 \sigma^2}{4} + O(\gamma^4).$$

Proof. Repeat the orthographic-coordinate calculation in Proposition 3.9 using the local singular coordinates $(0, \sigma, 0)$ at the critical pair (u_2, v_2) . In the notation of (28), this means $x = z = 0$, $y = \delta$, and $d = 1$. The squared area is one and the curvature is

$$-\frac{\delta^2 - 1 + \sqrt{1 - \delta^2}}{2(1 - \delta^2)} = -\frac{1 - \sqrt{1 - \delta^2}}{2\sqrt{1 - \delta^2}},$$

which is negative for $0 < |\delta| < 1$. Its Taylor expansion gives (31). $\square$

Corollary 3.11 (Four-dimensional detection and complete isolation). *Let $m, n \geq 2$. For every nonzero $A : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{m+1}$, choose an ordered singular frame and set $N_{123} = N_{\{1,2,3\}}$. For every nonzero Riemannian parameter, $N_{123} \cong \mathbb{S}^2 \times \mathbb{S}^2$ contains a plane of strictly negative ambient sectional curvature. Consequently,*

$$\left\{ \gamma : g_{\gamma, A} \text{ is Riemannian and } \sec_{g_{\gamma, A}} \geq 0 \right\} = \{0\}.$$

Proof. If $\sigma_2 > 0$, Proposition 3.9 supplies a negative oblique plane in N_{123} . If $\sigma_2 = 0$, then A has rank one and Proposition 3.10 supplies a negative oblique plane in the same submanifold. At $\gamma = 0$ the metric is the nonnegatively curved round product. $\square$

Corollary 3.12 (Explicit dimension-free negative margin). *Fix $0 < \epsilon < 1/2$, and let $r_\epsilon \in (0, 1)$ be the unique solution of*

$$(32) \quad (1 - r_\epsilon)^2 = \epsilon r_\epsilon^3.$$

Set

$$c_\epsilon = \frac{\epsilon^4 r_\epsilon^3}{16} > \frac{\epsilon^4}{128}.$$

For every $m, n \geq 2$, every nonzero A , and every γ satisfying

$$\epsilon \leq |\gamma|(\sigma_1(A) + \sigma_2(A)) \leq 1 - \epsilon,$$

one has

$$\min_{\Pi \in \text{Gr}_2(T(\mathbb{S}^m \times \mathbb{S}^n))} \sec_{g_\gamma, A}(\Pi) \leq -c_\epsilon.$$

Proof. The antipodal isometry lets us assume $\gamma > 0$. Put

$$x = \gamma\sigma_1, \quad y = \gamma\sigma_2, \quad z = \gamma\sigma_3, \quad p = x + y, \quad q = \frac{y}{p}.$$

Thus $0 \leq q \leq 1/2$ and $\epsilon \leq p \leq 1 - \epsilon$. When $x > y$, Theorem 3.5, together with $z \leq y$ and the fact that its last denominator is at most one, gives a plane Π_{gap} with

$$(33) \quad \begin{aligned} -\sec(\Pi_{\text{gap}}) &\geq \frac{(x^2 - y^2)^3}{16(x^2 + y^2)} \\ &\geq \frac{\epsilon^4(1 - 2q)^3}{16}. \end{aligned}$$

When $x = y$, define the first lower bound to be zero; only the oblique estimate below is then used.

For the oblique plane in Proposition 3.9, its proof gives

$$\mathcal{N}(x, y, z) \geq \min\{x, 1 - x\}(y^2 + z^2).$$

Moreover $\Delta_2 \Delta_3 \leq (1 - x)^4 \leq 1$, $x \geq p/2 \geq \epsilon/2$, and $1 - x \geq 1 - p \geq \epsilon$. Therefore

$$(34) \quad -\sec(\Pi_{\text{obl}}) \geq \frac{\epsilon^3 q^2}{4}.$$

Writing $r = 1 - 2q$, the larger of (33) and (34) is at least

$$\frac{1}{16} \max\{\epsilon^4 r^3, \epsilon^3(1 - r)^2\}.$$

The increasing and decreasing terms meet exactly at the unique root in (32); hence their maximum is at least c_ϵ . Finally $r_\epsilon > 1/2$, because at $r = 1/2$ the two sides of (32) are respectively $1/4$ and $\epsilon/8$. This proves the simpler displayed bound as well. $\square$

The following specialization uses a different plane than the canonical choice in Theorem 3.5 and gives a particularly simple curvature function.

Corollary 3.13 (A simple zero-threshold specialization). *On $\mathbb{S}^2 \times \mathbb{S}^2$, let*

$$A_0 = \begin{pmatrix} 0 & 2 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

At $p = (e_0, e_0)$ take the background-mixed plane spanned by

$$\mathbf{X} = ((0, 1, 0), 0), \quad \mathbf{Y} = (0, (0, 1, 0)).$$

For every

$$0 < |\gamma| < \frac{1}{\sqrt{5} + 1},$$

the metric g_{γ, A_0} is Riemannian on the whole product and

$$(35) \quad \sec_{g_{\gamma, A_0}}(\mathbf{X} \wedge \mathbf{Y}) = -\frac{3\gamma^4}{4(1 - \gamma^2)} < 0.$$

Thus no matrix-independent positive interval of mixed-curvature nonnegativity exists, even after normalizing by the sharp Riemannian endpoint.

Proof. Since $A_0^\top A_0 = \text{diag}(1, 5, 0)$, the Riemannian interval follows from Proposition 3.2. At p one has

$$a = (1, 0), \quad b = (2, 0), \quad P_x A_0 P_y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

In the ordered tangent basis the metric and the three covectors from Theorem 3.4 are

$$g_{\gamma, A_0} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & \gamma & 0 \\ 0 & \gamma & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

$$\eta_{XX} = \eta_{YY} = (-1/2, 0, -1, 0), \quad \eta_{XY} = (-1, 0, -1/2, 0).$$

Formula (13) now reduces to (35); the squared area is one. $\square$

A different ray illustrates that an individual mixed plane can have a nonzero positive-to-negative crossing, even though Theorem 3.5 already supplies another plane with negative curvature along the punctured ray.

Proposition 3.14 (A plane-wise nonzero sign-change threshold). *Let*

$$B = \frac{1}{100} \begin{pmatrix} 1 & 2 & -32 \\ -48 & -23 & 3 \\ -39 & -15 & -3 \end{pmatrix}, \quad \gamma_{\mathbb{R}} = \frac{1}{\sigma_1(B) + \sigma_2(B)}.$$

At $p = (e_0, e_0)$ consider the plane spanned by

$$\mathbf{X} = ((0, 63/100, 78/100), 0), \quad \mathbf{Y} = (0, (0, 9/100, -1)).$$

There is a unique $\gamma_* \in (0, \gamma_R)$ at which its sectional curvature changes sign, and

$$(36) \quad \frac{4679671}{5000000} < \gamma_* < \frac{9359343}{10000000}, \quad \gamma_* = 0.935934207005572 \dots$$

The curvature is positive for $0 < \gamma < \gamma_*$, zero at γ_* , and negative for $\gamma_* < \gamma < \gamma_R$.

Proof. Exact substitution in (13) gives

$$(37) \quad \sec_{g_{\gamma,B}}(X \wedge Y) = \frac{11\gamma^2 P(\gamma)}{4Q(\gamma)L(\gamma)},$$

where

$$\begin{aligned} P(\gamma) &= 1873383134707\gamma^3 - 187783301783700\gamma^2 \\ &\quad - 6938955210000\gamma + 169451579000000, \\ Q(\gamma) &= 28151192500 - 563023850\gamma - 7287743\gamma^2, \\ L(\gamma) &= 489\gamma^4 + 6160\gamma^3 - 306400\gamma^2 - 160000\gamma + 4000000. \end{aligned}$$

On the Riemannian interval, Q and L are positive: in fact, up to positive constants they are respectively the squared area of this plane and the determinant of $g_{\gamma,B}$ at p .

The crossing lies strictly inside the global Riemannian interval. Indeed,

$$\left(\frac{47}{50}\right)^2 2\|B\|_F^2 = \frac{6213917}{6250000} < 1,$$

so $\gamma_R > 47/50$. On the other hand, $\sigma_1(B) + \sigma_2(B) \geq \|Be_0\| = \sqrt{3826}/100 > 1/2$, so $\gamma_R < 2$.

On $[0, 2]$ the cubic P is strictly decreasing. Indeed,

$$\begin{aligned} P''(\gamma) &= 11240298808242\gamma - 375566603567400 < 0, \\ P'(0) &= -6938955210000 < 0. \end{aligned}$$

Finally, exact rational substitution gives

$$P(4679671/5000000) > 0, \quad P(9359343/10000000) < 0.$$

Thus P has exactly one zero in the Riemannian interval, with the bracket (36); the sign statement follows from (37). $\square$

Remark 3.15. Corollary 3.13 shows that the nonnegative γ^2 coefficient in (14) is not a uniform curvature barrier. When X is parallel to a and Y is parallel to b , that coefficient vanishes and the first nonzero term can be negative; in (35) it is $-3\gamma^4/4 + O(\gamma^6)$.

4. FLAT TOTALLY GEODESIC TORI

Fix a singular value decomposition

$$(38) \quad Av_k = \sigma_k u_k, \quad 1 \leq k \leq q,$$

with orthonormal bases completed in the left and right null spaces if necessary.

For $i < j$, let

$$(39) \quad C_{ij} = \mathbb{S}^m \cap \text{span}\{u_i, u_j\}, \quad D_{ij} = \mathbb{S}^n \cap \text{span}\{v_i, v_j\}, \quad T_{ij} = C_{ij} \times D_{ij}.$$

Each T_{ij} is a smoothly embedded 2-torus.

Proposition 4.1. *Whenever $g_{\gamma,A}$ is Riemannian, every torus T_{ij} in (39) is totally geodesic for $g_{\gamma,A}$.*

Proof. This is Proposition 3.8 with $I = \{i, j\}$. The argument also covers the case where one of the paired great circles is the entire sphere. $\square$

Proposition 4.2. *The metric induced by $g_{\gamma,A}$ on every T_{ij} is flat.*

Proof. Parametrize the two great circles by

$$x(u) = \cos u u_i + \sin u u_j, \quad y(v) = \cos v v_i + \sin v v_j.$$

The restriction of h is $du^2 + dv^2$. Since T_{ij} is also totally geodesic for h , the ambient Hessian restricts to the intrinsic Hessian of

$$(40) \quad \phi(u, v) = s_A|_{T_{ij}} = \sigma_i \cos u \cos v + \sigma_j \sin u \sin v.$$

Introduce local coordinates

$$\alpha = u + v, \quad \beta = u - v,$$

and abbreviate

$$(41) \quad a = \frac{\sigma_i + \sigma_j}{2}, \quad b = \frac{\sigma_i - \sigma_j}{2}.$$

The elementary product-to-sum identities give

$$\phi = b \cos \alpha + a \cos \beta, \quad du^2 + dv^2 = \frac{1}{2}(d\alpha^2 + d\beta^2).$$

It follows that

$$(42) \quad g_{\gamma,A}|_{T_{ij}} = \left(\frac{1}{2} - \gamma b \cos \alpha\right) d\alpha^2 + \left(\frac{1}{2} - \gamma a \cos \beta\right) d\beta^2.$$

Both coefficients are positive because the ambient metric is positive definite. Each depends on only its own coordinate. Locally making the changes of variables

$$\hat{\alpha} = \int_0^\alpha \sqrt{\frac{1}{2} - \gamma b \cos t} dt, \quad \hat{\beta} = \int_0^\beta \sqrt{\frac{1}{2} - \gamma a \cos t} dt$$

turns (42) into $d\hat{\alpha}^2 + d\hat{\beta}^2$. Hence its Gaussian curvature vanishes identically. $\square$

Proof of Theorem 1.1. Proposition 3.2 gives the exact Riemannian range. By Propositions 4.1 and 4.2, every chosen singular frame supplies $\binom{q}{2}$ flat totally geodesic tori. The Gauss equation for a totally geodesic surface identifies its intrinsic Gaussian curvature with the ambient sectional curvature of its tangent plane. That curvature is therefore zero at every point of every T_{ij} , which rules out strictly positive sectional curvature. When $m, n \geq 2$ and $\sigma_2 > 0$, Proposition 3.9 supplies strict negative curvature on the totally geodesic singular $\mathbb{S}^2 \times \mathbb{S}^2$. In the remaining rank-one case, Proposition 3.10 supplies strict negative curvature instead. The assertion for $A = 0$ is immediate from the definition. $\square$

Remark 4.3 (Repeated and zero singular values). The theorem does not require A to have full rank or simple singular values. When a positive singular value is repeated, the choice of singular frame is nonunique and may produce continuous families of flat tori. When some $\sigma_j = 0$, the same construction still works: the corresponding paired great circles are fixed by the involution and the separated formula (42) remains valid. The count $\binom{q}{2}$ is therefore a guaranteed count for each chosen frame, not a classification of all flat tori.

5. INHERITED SYMMETRY AND THE FULL ISOMETRY GROUP

The flat-torus proof uses finite involutions, but it is useful to state exactly how much of the background product symmetry is preserved. Let

$$(43) \quad G_A = \{(R, S) \in O(m+1) \times O(n+1) : R^\top A S = A\}.$$

This is the stabilizer of A for the natural left-right action.

Proposition 5.1. *For $\gamma \neq 0$, the factor-preserving isometries of the round product which remain isometries of $g_{\gamma, A}$ are exactly G_A . If the distinct positive singular values of A have multiplicities $d_1, \dots, d_k$ and $r_A = \text{rank } A$, then*

$$(44) \quad G_A \cong \left(\prod_{\ell=1}^k O(d_\ell)_{\text{diag}} \right) \times O(m+1-r_A) \times O(n+1-r_A).$$

Here $O(d_\ell)_{\text{diag}}$ acts by the same orthogonal matrix on the corresponding left and right singular subspaces.

Proof. If $(R, S) \in G_A$, then

$$s_A(Rx, Sy) = \langle x, R^\top A S y \rangle = s_A(x, y).$$

The associated product isometry therefore preserves h and $\text{Hess}_h s_A$, so it preserves $g_{\gamma, A}$.

Conversely, suppose the product isometry associated to (R, S) preserves $g_{\gamma, A}$. Since it already preserves h and $\gamma \neq 0$, the function

$$f = s_{R^\top A S} - s_A$$

satisfies $\text{Hess}_h f = 0$. On a compact connected manifold such a function is constant: at a maximum its gradient vanishes, while $\text{Hess} f = 0$ makes the gradient parallel. Both bilinear functions have zero integral over the product of spheres, so the constant is zero. Thus $R^\top AS = A$.

Finally, write A in singular-value block form. The relation $R^\top AS = A$ forces R and S to preserve the eigenspaces of $AA^\top$ and $A^\top A$. On a positive singular block of value ρ , the relation is $R_\rho^\top(\rho I)S_\rho = \rho I$, hence $R_\rho = S_\rho$. On the two null spaces the orthogonal actions are independent. This gives (44). $\square$

Corollary 5.2 (Curvature mechanisms and symmetry). *Assume $m, n \geq 2$, $A \neq 0$, and $\gamma \neq 0$ is in the Riemannian interval. Then one of the following holds:*

- (1) $\sigma_1 > \sigma_2$, and $g_{\gamma,A}$ has a background-mixed plane of strictly negative sectional curvature;
- (2) $\sigma_1 = \sigma_2$, and $g_{\gamma,A}$ has a negative oblique plane in a totally geodesic singular $\mathbb{S}^2 \times \mathbb{S}^2$, while the inherited factor-preserving isometry group G_A contains a circle subgroup.

Proof. The first alternative is Theorem 3.5. In the second, Proposition 3.9 gives the negative plane. The top positive singular value has multiplicity at least two, so Proposition 5.1 also exhibits a diagonal $O(d)$ factor with $d \geq 2$. Its identity component contains a circle. $\square$

Corollary 5.3 (The generic $\mathbb{S}^2 \times \mathbb{S}^2$ case). *If $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is invertible with three distinct singular values, then the inherited factor-preserving group is*

$$G_A \cong O(1)^3 \cong (\mathbb{Z}/2\mathbb{Z})^3.$$

It contains the three involutions whose fixed sets are the tori T_{12} , T_{13} , and T_{23} , but it contains no circle subgroup.

Remark 5.4 (What is not being claimed). Proposition 5.1 computes the intersection of $\text{Isom}(M, g_{\gamma,A})$ with the factor-preserving isometry group of the round product. It does not classify the full isometry group of $g_{\gamma,A}$. For example, equal-dimensional factors can admit a factor interchange under an additional algebraic condition, and an exceptional parameter could in principle have isometries not inherited from h . Accordingly, the finite group in the corollary is not asserted to be the full isometry group. The curvature obstruction does not require such an assertion.

6. A TWO-JET REPAIR CRITERION

The exact rays above have a vanishing first-order curvature signal for a structural reason:

$$(45) \quad \text{Hess}_h f = \frac{1}{2} \mathcal{L}_{\nabla f} h.$$

Thus their first metric variation is tangent to the diffeomorphism orbit of h . The next statement separates that gauge term from the first curvature-relevant correction. It applies to an arbitrary Riemannian product; we then specialize it to s_A .

Let $h = h_1 \oplus h_2$ on $M_1 \times M_2$, let ∇ be its connection, and let B be a symmetric 2-tensor. For an h -unit mixed pair $X \in TM_1$, $Y \in TM_2$, define

$$(46) \quad \begin{aligned} \mathbf{m}_B(X, Y) = & \frac{1}{2}((\nabla_{X,Y}^2 B)(X, Y) + (\nabla_{Y,X}^2 B)(X, Y) \\ & - (\nabla_{X,X}^2 B)(Y, Y) - (\nabla_{Y,Y}^2 B)(X, X)). \end{aligned}$$

Because $R^h(X, Y) = 0$, the two cross derivatives agree, but the symmetrized form makes the curvature linearization transparent.

Theorem 6.1 (Two-jet normal form and mixed repair condition). *Let g_t be a smooth family of metrics on $M_1 \times M_2$ with*

$$(47) \quad g_t = h + t \operatorname{Hess}_h f + t^2 B + O(t^3)$$

in C^2 . Let Φ_t be the local flow of $-\frac{1}{2}\nabla f$. Then

$$(48) \quad \Phi_t^* g_t = h + t^2 \hat{B} + O(t^3), \quad \hat{B} = B - \frac{1}{8} \mathcal{L}_{\nabla f}^2 h.$$

For every $p \in M_1 \times M_2$ and every h -unit mixed pair X, Y at p ,

$$(49) \quad \sec_{g_t}(d\Phi_t(X \wedge Y)) = t^2 \mathbf{m}_{\hat{B}}(X, Y) + O(t^3).$$

Consequently, a necessary condition for g_t to have nonnegative sectional curvature for all sufficiently small t is

$$(50) \quad \mathbf{m}_{\hat{B}}(X, Y) \geq 0 \quad \text{for every background-mixed unit pair } X, Y.$$

If the inequality is strict in the opposite direction for one pair, g_t has negative sectional curvature for every sufficiently small nonzero t .

Proof. Put $Z = \nabla f$. Pullback by the flow of $-Z/2$ acts to second order as $\exp(-\frac{t}{2}\mathcal{L}_Z)$. Using (45) in (47), the coefficient of t cancels, while the coefficient of t^2 is

$$B - \frac{1}{2} \mathcal{L}_Z \operatorname{Hess} f + \frac{1}{8} \mathcal{L}_Z^2 h = B - \frac{1}{8} \mathcal{L}_Z^2 h.$$

This proves (48).

Choose product h -normal coordinates at p whose first coordinate directions include X and Y . The lower-index curvature formula for $h + t^2 \hat{B} + O(t^3)$, evaluated on (X, Y, X, Y) , has coefficient

$$\frac{1}{2}((\nabla_{X,Y}^2 \hat{B})(X, Y) + (\nabla_{Y,X}^2 \hat{B})(X, Y) - (\nabla_{X,X}^2 \hat{B})(Y, Y) - (\nabla_{Y,Y}^2 \hat{B})(X, X)).$$

All background curvature terms with both factor directions vanish, as do the Christoffel-quadratic terms at this order. The squared area of the plane is $1 + O(t^2)$, so division gives (49). Sectional curvature is preserved by pullback, and the sign conclusions follow. $\square$

On a closed product the inequalities in (50) satisfy a global compatibility identity. The functional (46) is the classical mixed-plane curvature linearization [4]; tracing it gives the following global consequence in the present gauge-normalized setting. Let $e_1, \dots, e_m$ and $f_1, \dots, f_n$ be local product orthonormal frames, and write $\text{tr}_1 B$ and $\text{tr}_2 B$ for the partial traces of B .

Theorem 6.2 (Integrated mixed two-jet rigidity). *If M_1 and M_2 are closed, then every symmetric 2-tensor B satisfies*

$$(51) \quad \int_{M_1 \times M_2} \sum_{i=1}^m \sum_{\alpha=1}^n \mathfrak{m}_B(e_i, f_\alpha) dV_h = 0.$$

Consequently, if $\mathfrak{m}_B(X, Y) \geq 0$ for every background-mixed unit pair, then

$$(52) \quad \mathfrak{m}_B(X, Y) = 0 \quad \text{for every background-mixed pair } X, Y.$$

In particular, any family in Theorem 6.1 which has nonnegative sectional curvature for all sufficiently small t must satisfy

$$(53) \quad \mathfrak{m}_{\widehat{B}}(X, Y) = 0 \quad \text{for every background-mixed pair } X, Y.$$

Thus every such mixed curvature is $O(t^3)$ after gauge normalization; none can be lifted strictly at quadratic order without another becoming negative.

Proof. Tracing (46) in product normal frames gives

$$(54) \quad \sum_{i,\alpha} \mathfrak{m}_B(e_i, f_\alpha) = \text{div}_1 \text{div}_2(B|_{TM_1 \otimes TM_2}) - \frac{1}{2} \Delta_1(\text{tr}_2 B) - \frac{1}{2} \Delta_2(\text{tr}_1 B).$$

The two possible orders in the first term agree because the mixed curvature of the product connection vanishes. Each term on the right-hand side is a divergence. Its integral is zero on the closed product, proving (51).

Now suppose every mixed value is nonnegative. Its trace in any product orthonormal frame is a nonnegative function with zero integral, hence vanishes identically. Every summand in that frame is therefore zero. At a given point any prescribed mixed unit pair can be included in such frames, which proves (52). The last assertion follows from Theorem 6.1. $\square$

The global test is therefore feasible, but only at equality. Put $Z = \nabla f$, let Ψ_t be the flow of $Z/2$, and set

$$(55) \quad B_{\text{gauge}} = \frac{1}{8} \mathcal{L}_Z^2 h.$$

Then

$$\Psi_t^* h = h + t \text{Hess } f + t^2 B_{\text{gauge}} + O(t^3), \quad \widehat{B} = 0.$$

This correction solves all the two-jet equalities, but produces only a family isometric to the original product. The kernel is not purely gauge: tensors pulled back from the individual factors also satisfy (52), as expected from ordinary product variations. Theorem 6.4 below classifies the full kernel modulo these two trivial sectors. Testing its first nonzero higher-order

curvature is a separate problem. The higher-order positive mixed variations constructed in [4] show why this second step is not vacuous; they also illustrate that improving all mixed planes does not by itself produce positive sectional curvature on every plane.

Condition (53) is necessary, not sufficient: when its coefficient vanishes, higher jets can decide the sign. It is invariant under the remaining second-order gauge freedom. Indeed, replacing $\widehat{B}$ by $\widehat{B} + \mathcal{L}_W h$ changes the linearized curvature by a Lie derivative of the product curvature tensor, which vanishes on a mixed four-tuple.

For the bilinear potentials, the abstract condition gives explicit values which any proposed nonlinear repair must attain.

Corollary 6.3 (Explicit two-jet cost for bilinear Hessian paths). *Assume $m, n \geq 2$, let $A \neq 0$, and consider*

$$(56) \quad g_t = h + t \operatorname{Hess}_h s_A + t^2 B + O(t^3).$$

Suppose g_t has nonnegative sectional curvature for all sufficiently small t . Choose an ordered singular frame and pad the singular values by zeros.

If $\sigma_2 > 0$, set

$$p = (u_1, v_1), \quad X = (u_3, 0), \quad Y = (0, v_2).$$

Then nonnegative curvature forces the exact repair identity

$$(57) \quad \mathfrak{m}_B(X, Y) = \frac{\sigma_2^2 + \sigma_3^2}{4}.$$

If A has rank one, set instead

$$p = (u_2, v_2), \quad X = (u_3, 0), \quad Y = (0, v_1).$$

Then nonnegative curvature forces

$$(58) \quad \mathfrak{m}_B(X, Y) = \frac{\sigma_1^2}{4}.$$

In particular, $B = 0$ violates one of these identities for every $A \neq 0$. Hence every nontrivial exact bilinear Hessian ray is already obstructed by its two-jet modulo diffeomorphisms.

Proof. First suppose $\sigma_2 > 0$. At $p = (u_1, v_1)$ the gradient of s_A vanishes, and the Hessian endomorphism has eigenvectors

$$e_i^\pm = \frac{(u_i, \pm v_i)}{\sqrt{2}}, \quad \operatorname{Hess} s_A(e_i^\pm) = (-\sigma_1 \pm \sigma_i) e_i^\pm.$$

Consequently the differential of the flow in Theorem 6.1, applied to $Y = (0, v_2)$ and $X = (u_3, 0)$, agrees to first order with the metric-normalized vectors in (26). More explicitly,

$$d\Phi_t(e_i^\pm) = \left(1 + \frac{t}{2}(\sigma_1 \mp \sigma_i)\right) e_i^\pm + O(t^2) = (1 - t\sigma_1 \pm t\sigma_i)^{-1/2} e_i^\pm + O(t^2).$$

Expanding (27) with $(x, y, z) = t(\sigma_1, \sigma_2, \sigma_3)$ gives

$$(59) \quad \sec = -\frac{t^2}{4}(\sigma_2^2 + \sigma_3^2) + O(t^3).$$

Theorem 6.1 therefore gives

$$\mathfrak{m}_{-\frac{1}{8}\mathcal{L}_{\nabla_{s_A}}^2} h(X, Y) = -\frac{\sigma_2^2 + \sigma_3^2}{4}.$$

Hence

$$\mathfrak{m}_{\widehat{B}}(X, Y) = \mathfrak{m}_B(X, Y) - \frac{\sigma_2^2 + \sigma_3^2}{4}.$$

Theorem 6.2 forces this value to vanish, which proves (57).

If A has rank one, the same comparison at (u_2, v_2) uses the vectors in Proposition 3.10. Equation (31) gives

$$\mathfrak{m}_{-\frac{1}{8}\mathcal{L}_{\nabla_{s_A}}^2} h(X, Y) = -\frac{\sigma_1^2}{4},$$

so $\mathfrak{m}_{\widehat{B}}(X, Y) = \mathfrak{m}_B(X, Y) - \sigma_1^2/4$. Again Theorem 6.2 forces this value to vanish, proving (58). The right-hand side is positive in both cases. $\square$

6.1. The full coupled kernel modulo gauge and product tensors. The partial-divergence reduction below goes back to Bourguignon–Deschamps–Sentenac [3]. Their analysis settles the case without Killing fields and, in the general case, records the coupled subspace generated by Killing one-forms. The tensor-product structure of the remaining equation identifies that subspace as the entire kernel.

Assume throughout this subsection that M_1 and M_2 are closed and connected. Let

$$\mathcal{D}_i : \Omega^1(M_i) \longrightarrow \Gamma(S^2 T^* M_i), \quad \mathcal{D}_i \alpha = \frac{1}{2} \mathcal{L}_{\alpha^\sharp} h_i$$

be the Killing operator, and put

$$\mathcal{K}_i = \ker \mathcal{D}_i, \quad \mathcal{Z}_i = \ker d_i^* \subset \Omega^1(M_i).$$

Thus $\mathcal{K}_i$ is the finite-dimensional space of Killing one-forms, $\mathcal{Z}_i$ is the space of co-closed one-forms, and $\mathcal{K}_i \subset \mathcal{Z}_i$. All orthogonal complements below are taken in L^2 . For a mixed covariant tensor $C \in \Gamma(T^* M_1 \boxtimes T^* M_2)$, let B_C denote the symmetric tensor whose pure blocks vanish and whose mixed block is

$$(60) \quad B_C((X, Y), (X', Y')) = C(X, Y') + C(X', Y).$$

Finally, define

$$\begin{aligned} \mathfrak{K}_h &= \{B : \mathfrak{m}_B(X, Y) = 0 \text{ for every mixed pair } X, Y\}, \\ \mathfrak{G}_h &= \{\mathcal{L}_W h : W \in \Gamma(TM)\}, \\ \mathfrak{P}_h &= \{\pi_1^* P + \pi_2^* Q : P \in \Gamma(S^2 T^* M_1), Q \in \Gamma(S^2 T^* M_2)\}. \end{aligned}$$

Theorem 6.4 (Full coupled-kernel classification). *Every element of $\mathfrak{K}_h$ has a decomposition*

$$(61) \quad B = \mathcal{L}_W h + \pi_1^* P + \pi_2^* Q + B_C,$$

where

$$(62) \quad C \in \mathfrak{H}_{12} := (\mathcal{K}_1 \otimes \mathcal{Z}_2) + (\mathcal{Z}_1 \otimes \mathcal{K}_2).$$

The tensor B_C is uniquely determined by the class of B modulo $\mathfrak{G}_h + \mathfrak{P}_h$. Equivalently,

$$(63) \quad \frac{\mathfrak{K}_h}{\mathfrak{G}_h + \mathfrak{P}_h} \cong \mathfrak{H}_{12}.$$

More precisely, the decomposition

$$(64) \quad \mathfrak{H}_{12} = (\mathcal{K}_1 \otimes \mathcal{K}_2) \oplus (\mathcal{K}_1 \otimes (\mathcal{Z}_2 \ominus \mathcal{K}_2)) \oplus ((\mathcal{Z}_1 \ominus \mathcal{K}_1) \otimes \mathcal{K}_2)$$

is L^2 -orthogonal. In particular, every nontrivial coupled class has a Killing leg on at least one factor, and no class with both legs orthogonal to the Killing fields survives.

Proof. Both $\mathfrak{G}_h$ and $\mathfrak{P}_h$ lie in $\mathfrak{K}_h$. For the first space this follows either from diffeomorphism invariance or by linearizing the product curvature tensor on a mixed four-tuple; for the second it follows by varying the two factor metrics separately.

Write an arbitrary symmetric tensor in block form as

$$B = P + B_C + Q,$$

where P and Q are families of pure symmetric tensors on the first and second factors. Apply the usual divergence–Killing-operator decomposition on each factor, with the other variable as a parameter. After subtracting a Lie derivative, we may suppose

$$(65) \quad \delta_1^{(2)} P = 0, \quad \delta_2^{(2)} Q = 0,$$

where $\delta_i^{(2)} = \mathcal{D}_i^*$ is the divergence on symmetric 2-tensors. The elliptic solution can be chosen orthogonal to $\mathcal{K}_i$, so it depends smoothly on the parameter from the other factor.

Polarizing $\mathfrak{m}_B(X, Y)$ separately in X and Y gives a section $\mathcal{M}(B)$ of $S^2 T^* M_1 \boxtimes S^2 T^* M_2$. In the block notation above,

$$(66) \quad \mathcal{M}(B) = \mathcal{D}_1 \mathcal{D}_2 C - \frac{1}{2} \text{Hess}_1 Q - \frac{1}{2} \text{Hess}_2 P.$$

Here, for example, $(\text{Hess}_1 Q)(X, X; Y, Y) = (\nabla_{X, X}^2 Q)(Y, Y)$. In adapted indices this is the pointwise identity

$$\mathcal{M}(B)_{ij\alpha\beta} = \nabla_{(i} \nabla_{(\alpha} C_{j)\beta)} - \frac{1}{2} \nabla_i \nabla_j Q_{\alpha\beta} - \frac{1}{2} \nabla_\alpha \nabla_\beta P_{ij},$$

where the two pairs of parentheses are symmetrized independently. Because product derivatives commute, (65) implies that $\text{Hess}_1 Q$ is in $\ker \delta_2^{(2)}$ in its second-factor slots. The other two terms in (66) are in $\text{im } \mathcal{D}_2$: the last one

is the second-factor Hessian of the scalar $P(X, X)$. Orthogonality of $\text{im } \mathcal{D}_2$ and $\ker \delta_2^{(2)}$ therefore gives $\text{Hess}_1 Q = 0$. Interchanging the factors then gives $\text{Hess}_2 P = 0$, and hence

$$(67) \quad \mathcal{D}_1 \mathcal{D}_2 C = 0.$$

On a closed connected manifold a function with zero Hessian is constant. Applied to the scalar coefficients of P and Q in the opposite factor, this shows that P and Q are pulled back from the individual factors. They may therefore be absorbed into $\mathfrak{P}_h$.

It remains to solve (67). Let $L_i = \mathcal{D}_i^* \mathcal{D}_i$. This is a nonnegative self-adjoint elliptic operator on one-forms, with kernel $\mathcal{K}_i$. Choose smooth orthonormal eigenbases $\{\alpha_r\}$ and $\{\beta_s\}$, with eigenvalues λ_r and μ_s , and expand $C = \sum c_{rs} \alpha_r \otimes \beta_s$. Then

$$(68) \quad \|\mathcal{D}_1 \mathcal{D}_2 C\|_{L^2}^2 = \sum_{r,s} \lambda_r \mu_s |c_{rs}|^2.$$

Consequently $c_{rs} = 0$ whenever both eigenvalues are positive, and

$$(69) \quad C \in (\mathcal{K}_1 \otimes \Omega^1(M_2)) + (\Omega^1(M_1) \otimes \mathcal{K}_2).$$

The smooth spectral projections onto the finite-dimensional Killing spaces show that the sums in (69) may be taken to be finite in the Killing index.

Writing Π_i for the L^2 projection onto $\mathcal{K}_i$, choose the nonredundant decomposition $C = (\Pi_1 \otimes I)C + ((I - \Pi_1) \otimes \Pi_2)C$. Use the Hodge splittings $\Omega^1(M_i) = dC^\infty(M_i) \oplus \mathcal{Z}_i$ on the non-Killing coefficients. If $\kappa \in \mathcal{K}_1$, $\lambda \in \mathcal{K}_2$, and u, v are smooth functions, then

$$(70) \quad B_{\kappa \otimes du} = \mathcal{L}_{u\kappa^\#} h, \quad B_{dv \otimes \lambda} = \mathcal{L}_{v\lambda^\#} h.$$

Thus every class has a representative in (62).

Such a representative is trace-free and divergence-free: in each summand one leg is Killing and the other is co-closed. Hence B_C is L^2 -orthogonal to every Lie derivative. It is also pointwise orthogonal to every pure product tensor. Therefore no nonzero B_C in $\mathfrak{H}_{12}$ belongs to $\mathfrak{G}_h + \mathfrak{P}_h$, proving uniqueness and (63). Finally, $(\mathcal{K}_1 \otimes \mathcal{Z}_2) \cap (\mathcal{Z}_1 \otimes \mathcal{K}_2) = \mathcal{K}_1 \otimes \mathcal{K}_2$, which gives the orthogonal splitting (64). $\square$

Corollary 6.5 (The sphere-product kernel). *Let $\mathfrak{k}_m \subset \Omega^1(\mathbb{S}^m)$ be the metric duals of the rotational fields $x \mapsto Ax$, $A \in \mathfrak{so}(m+1)$, and put*

$$\mathcal{Z}_m^{\geq 2} = \ker d^* \ominus \mathfrak{k}_m.$$

Then the coupled two-jet kernel on $\mathbb{S}^m \times \mathbb{S}^n$, modulo diffeomorphisms and factor tensors, is exactly

$$(71) \quad (\mathfrak{k}_m \otimes \mathfrak{k}_n) \oplus (\mathfrak{k}_m \otimes \mathcal{Z}_n^{\geq 2}) \oplus (\mathcal{Z}_m^{\geq 2} \otimes \mathfrak{k}_n).$$

For $\mathbb{S}^2 \times \mathbb{S}^2$, in coexact vector spherical harmonics, the allowed bidegrees are precisely

$$(72) \quad (1, 1), \quad (1, \ell) \ (\ell \geq 2), \quad (\ell, 1) \ (\ell \geq 2).$$

With the cutoff $\ell \leq L$ on both factors, the quotient therefore has dimension

$$(73) \quad 9 + 6 \sum_{\ell=2}^L (2\ell + 1) = 6(L + 1)^2 - 15.$$

Proof. Every Killing field on a round sphere is rotational, and $\dim \mathfrak{k}_m = \binom{m+1}{2}$. Thus Theorem 6.4 gives (71). On $\mathbb{S}^2$, the Hodge star writes every co-closed one-form as $*du$ modulo a constant potential. Expanding u in scalar spherical harmonics, degree 1 gives exactly the three rotational Killing forms, while degree ℓ has multiplicity $2\ell + 1$. This proves (72), and $\sum_{\ell=2}^L (2\ell + 1) = (L + 1)^2 - 4$, proving the dimension formula. $\square$

Thus the full quadratic search space is infinite-dimensional, but it is spectrally thin: one of the two harmonic legs must be rotational. The rotational tensor (95) below lies in the finite 9-dimensional $(1, 1)$ block. The next theorem tests the two one-sided towers $(1, \ell)$ and $(\ell, 1)$, $\ell \geq 2$, level by level and without a separability assumption.

6.2. Exact nonlinear tests in the coupled kernel. Nearby moving planes already appear in the classical higher-order analysis of Bourguignon–Deschamps–Sentenac [4, Props. 5.7 and 5.9], where separated Killing variations and a related field-valued class are obstructed. Here the coupled-kernel classification, a two-curl coefficient, and the spherical spectral gap make the sign explicit for arbitrary sums in the full quotient.

Identify the Killing one-forms on the first $\mathbb{S}^2$ with $\mathbb{R}^3$ by $a \mapsto \kappa_a$, where κ_a is dual to $x \mapsto a \times x$. For an $\mathbb{R}^3$ -valued one-form E on the second sphere, define

$$(74) \quad C_E(X, Y) = \langle x \times X, E_y(Y) \rangle.$$

Thus B_{C_E} is an arbitrary element of $\mathfrak{k}_2 \otimes \Omega^1(\mathbb{S}^2)$ in the notation of Corollary 6.5.

Lemma 6.6 (The moving-plane coefficient). *Fix $y \in \mathbb{S}^2$, choose a unit vector $x \perp \text{im } E_y$, and let $v \in x^\perp$ be unit. Choose unit $X \in T_x \mathbb{S}^2$ so that $x \times X = v$. For an oriented orthonormal pair $Y, Z \in T_y \mathbb{S}^2$, set*

$$\theta = \langle v, E \rangle, \quad f = *d\theta,$$

where the star is on the second sphere. For $c \in \mathbb{R}$, let

$$(75) \quad \Pi_s(c) = \text{span}\{(X, scZ), (0, Y)\}.$$

Then, as $s \rightarrow 0$,

$$(76) \quad \sec_{h+sB_{C_E}} \Pi_s(c) = s^2 \left(c^2 - c(Yf) + \frac{1}{4}f^2 \right) + O(s^3).$$

In particular, the choice $c = (Yf)/2$ gives

$$(77) \quad \sec_{h+sB_{C_E}} \Pi_s \left(\frac{Yf}{2} \right) = \frac{s^2}{4} (f^2 - (Yf)^2) + O(s^3).$$

Proof. Use oriented product-normal coordinates at (x, y) with coordinate vectors X on the first factor and Y, Z on the second. The condition $x \perp \text{im } E_y$ says that the first-factor curl of the combined Killing one-form $C_E(\cdot, Y)$ vanishes at the base point. The fixed mixed-plane term is therefore $s^2 f^2/4$: it is the squared Christoffel term in (94), with the same calculation applied linearly to the sum of Killing legs. The round curvature of the second factor contributes $s^2 c^2$ when scZ is added to the first spanning vector. Finally, the term bilinear in the metric variation and the plane displacement is $-s^2 c(Yf)$. Equivalently, at the normal-coordinate origin write $\theta = \theta_1 dy^1 + \theta_2 dy^2$. The three terms are

$$\frac{s^2}{4}(\partial_1 \theta_2 - \partial_2 \theta_1)^2, \quad s^2 c^2, \quad -s^2 c \partial_1(\partial_1 \theta_2 - \partial_2 \theta_1).$$

The squared area is $1 + O(s^2)$, proving (76). Completing the square proves (77). $\square$

Let $\mathcal{Z}_\ell \subset \Omega^1(\mathbb{S}^2)$ denote the coexact one-form spherical harmonics of degree ℓ .

Theorem 6.7 (Every fixed one-sided level has a quadratic oblique obstruction). *For every $\ell \geq 2$ and every nonzero*

$$C \in \mathfrak{k}_2 \otimes \mathcal{Z}_\ell,$$

there are a point, a mixed plane at that point, a family of nearby planes Π_s , and a constant $\delta_C > 0$ such that

$$(78) \quad \sec_{h+sB_C}(\Pi_s) = -\delta_C s^2 + O(s^3).$$

Consequently $h + sB_C$ has negative sectional curvature for every sufficiently small nonzero s . The same conclusion holds for every nonzero $C \in \mathcal{Z}_\ell \otimes \mathfrak{k}_2$.

Proof. Write $C = C_E$ as in (74), with each component of E in $\mathcal{Z}_\ell$, and put

$$F = *dE : \mathbb{S}^2 \rightarrow \mathbb{R}^3.$$

The function F is a nonzero vector-valued spherical harmonic of degree ℓ . With $\lambda_\ell = \ell(\ell + 1)$,

$$(79) \quad \int_{\mathbb{S}^2} |dF|^2 = \lambda_\ell \int_{\mathbb{S}^2} |F|^2.$$

If $|dF_y|_{\text{op}} \leq |F(y)|$ at every point, then $|dF|^2 \leq 2|F|^2$ pointwise, contradicting (79) because $\lambda_\ell \geq 6 > 2$. Hence there are y and a unit Y such that

$$(80) \quad |dF_y(Y)| > |F(y)|.$$

Set $v = dF_y(Y)/|dF_y(Y)|$. Componentwise Hodge theory gives $E = \pm \lambda_\ell^{-1} * dF$, so $\text{im } E_y = \text{im } dF_y$. Choose a unit $x \perp \text{im } E_y$ and X with $x \times X = v$. For $\theta = \langle v, E \rangle$ and $f = *d\theta$, one has

$$f(y) = \langle v, F(y) \rangle, \quad Yf = \langle v, dF_y(Y) \rangle = |dF_y(Y)|.$$

Thus (80) implies $|Yf| > |f|$. Lemma 6.6, with $c = (Yf)/2$, gives (78) with

$$\delta_C = \frac{1}{4}((Yf)^2 - f^2) > 0.$$

Interchanging the two sphere factors proves the final assertion. $\square$

Lemma 6.8 (The two-curl moving-plane coefficient). *Let C represent an element of the coupled mixed two-jet kernel on $\mathbb{S}^2 \times \mathbb{S}^2$. Fix oriented orthonormal ambient frames*

$$(x, X, \bar{X}), \quad (y, Y, \bar{Y}), \quad \bar{X} = x \times X, \quad \bar{Y} = y \times Y.$$

For the two scalar one-forms obtained by freezing one argument of C , put

$$(81) \quad \alpha = *_1 d_1(C(\cdot, Y)), \quad \beta = *_2 d_2(C(X, \cdot)).$$

For $a, b \in \mathbb{R}$, let

$$(82) \quad \Pi_s(a, b) = \text{span}\{(X, sa\bar{Y}), (sb\bar{X}, Y)\}.$$

Then

$$(83) \quad \sec_{h+sB_C} \Pi_s(a, b) = s^2 \left(a^2 + b^2 - a(Y\beta) - b(X\alpha) + \frac{1}{4}(\alpha^2 + \beta^2) \right) + O(s^3).$$

In particular, $a = (Y\beta)/2$ and $b = (X\alpha)/2$ give

$$(84) \quad \sec_{h+sB_C} \Pi_s = -\frac{s^2}{4} \mathcal{D}_C + O(s^3), \quad \mathcal{D}_C := (X\alpha)^2 + (Y\beta)^2 - \alpha^2 - \beta^2.$$

Proof. Use product-normal coordinates adapted to the two displayed frames. Since C is in the mixed two-jet kernel, the term linear in s on the fixed mixed plane vanishes. The round curvatures of the two factors contribute $s^2 a^2$ and $s^2 b^2$. The terms bilinear in the metric variation and the two plane displacements are respectively

$$-s^2 a(Y\beta), \quad -s^2 b(X\alpha).$$

Finally, the first variation of the Christoffel symbol on the fixed mixed plane has orthogonal factor components $\alpha/2$ and $\beta/2$. Its squared norm contributes $s^2(\alpha^2 + \beta^2)/4$; the two pure Christoffel vectors lie in opposite factors, so their pairing vanishes. The squared area is $1 + O(s^2)$, so it does not change the quadratic coefficient. This proves (83); minimizing its two independent squares proves (84). $\square$

Theorem 6.9 (The full coupled kernel has a quadratic oblique obstruction). *Let $0 \neq C \in \mathfrak{H}_{12}$ on $\mathbb{S}^2 \times \mathbb{S}^2$, where $\mathfrak{H}_{12}$ is the full coupled kernel in (62). Then there are a point, a background-mixed plane at that point, a family of nearby planes Π_s , and a constant $\delta_C > 0$ such that*

$$(85) \quad \sec_{h+sB_C}(\Pi_s) = -\delta_C s^2 + O(s^3).$$

Consequently $h + sB_C$ has negative sectional curvature for every sufficiently small nonzero s .

Proof. The orthogonal harmonic splitting in (64) gives a unique decomposition

$$(86) \quad C = C_M + C_E + C_D,$$

where

$$\begin{aligned} C_M(X, Y) &= \langle x \times X, M(y \times Y) \rangle, \\ C_E(X, Y) &= \langle x \times X, E_y(Y) \rangle, \\ C_D(X, Y) &= \langle D_x(X), y \times Y \rangle. \end{aligned}$$

Here $M : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is linear, while the components of the $\mathbb{R}^3$ -valued coexact one-forms E and D are L^2 -orthogonal to the degree-one harmonics. Put

$$F = *dE : \mathbb{S}^2 \rightarrow \mathbb{R}^3, \quad G = *dD : \mathbb{S}^2 \rightarrow \mathbb{R}^3.$$

Average the defect $\mathcal{D}_C$ in (84) over the two oriented orthonormal frame bundles, using probability Haar measure, and write $\text{Av}_{\mathbb{S}^2}$ for normalized area average. The isotropic frame identities

$$\text{Av} \langle x, z \rangle^2 = \text{Av} \langle X, z \rangle^2 = \text{Av} \langle \bar{X}, z \rangle^2 = \frac{1}{3} |z|^2$$

and their second-factor analogues give

$$(87) \quad \begin{aligned} \text{Av} \mathcal{D}_C &= \frac{1}{3} \left\{ \frac{1}{2} \text{Av}_{\mathbb{S}^2} |dF|^2 - \text{Av}_{\mathbb{S}^2} |F|^2 \right. \\ &\quad \left. + \frac{1}{2} \text{Av}_{\mathbb{S}^2} |dG|^2 - \text{Av}_{\mathbb{S}^2} |G|^2 \right\}. \end{aligned}$$

For completeness, the two factorwise curls used here are

$$\begin{aligned} \alpha &= 2 \langle x, M\bar{Y} \rangle + 2 \langle x, E(Y) \rangle + \langle G(x), \bar{Y} \rangle, \\ \beta &= 2 \langle \bar{X}, My \rangle + \langle \bar{X}, F(y) \rangle + 2 \langle y, D(X) \rangle, \end{aligned}$$

and their relevant derivatives are

$$\begin{aligned} X\alpha &= 2 \langle X, M\bar{Y} \rangle + 2 \langle X, E(Y) \rangle + \langle dG_x(X), \bar{Y} \rangle, \\ Y\beta &= 2 \langle \bar{X}, MY \rangle + \langle \bar{X}, dF_y(Y) \rangle + 2 \langle Y, D(X) \rangle. \end{aligned}$$

These formulas also verify (87) directly. The M -square has zero average. Cross terms between different summands vanish: the higher coexact one-forms are orthogonal to the Killing one-forms, and their curls contain no degree-one scalar harmonics. The remaining pure E and D terms are exactly the two lines on the right-hand side of (87).

Every component of F and G is a sum of scalar spherical harmonics of degrees at least two. Hence the spherical spectral gap gives

$$\int |dF|^2 \geq 6 \int |F|^2, \quad \int |dG|^2 \geq 6 \int |G|^2.$$

Since $\mathbb{S}^2$ has no harmonic one-forms, $F = 0$ exactly when $E = 0$, and similarly for G and D . Thus (87) is strictly positive whenever $E \neq 0$ or $D \neq 0$. Some pair of frames then has $\mathcal{D}_C > 0$, and Lemma 6.8 gives (85).

It remains to treat $E = D = 0$ and $M \neq 0$. Choose oriented signed singular frames with $Mv_i = \tau_i u_i$ and $\tau_1 \neq 0$, and set

$$(x, X, \bar{X}) = (u_2, u_1, -u_3), \quad (y, Y, \bar{Y}) = (v_2, v_3, v_1).$$

Then $\alpha = \beta = 0$, while

$$X\alpha = 2\tau_1, \quad Y\beta = -2\tau_3.$$

Consequently $\mathcal{D}_{C_M} = 4(\tau_1^2 + \tau_3^2) > 0$, and the same lemma finishes the proof. $\square$

Corollary 6.10 (Every coupled kernel ray fails quartically). *Let $0 \neq C \in \mathfrak{H}_{12}$, let $\varphi \in C^\infty(\mathbb{S}^2 \times \mathbb{S}^2)$, put $W = \nabla\varphi$, and let Ψ_t be the flow of $W/2$. Then*

$$(88) \quad g_t = \Psi_t^*(h + t^2 B_C) = h + t \operatorname{Hess} \varphi + t^2 \left(\frac{1}{8} \mathcal{L}_W^2 h + B_C \right) + O(t^3)$$

passes every quadratic mixed repair identity, but contains planes with

$$(89) \quad \sec_{g_t} = -\delta_C t^4 + O(t^6) < 0$$

for all sufficiently small nonzero t .

Proof. The expansion is the same pullback calculation as in (55). The normalized second-order tensor is B_C , which belongs to the mixed two-jet kernel. Pullback preserves sectional curvature; substitute $s = t^2$ in (85). $\square$

Theorem 6.9 includes cross-degree superpositions, simultaneous mixtures of both one-sided towers, and arbitrary additions from the full $(1, 1)$ block. For comparison, separated Killing variations already occur in the classical variation theory [4]. We now record their exact fixed-mixed curvature formula and then pass to the canonical transverse sum in the $(1, 1)$ block. For one-forms ξ on M_1 and η on M_2 , define the pure mixed symmetric tensor $B_{\xi, \eta}$ by

$$(90) \quad B_{\xi, \eta}((X, Y), (X', Y')) = \xi(X)\eta(Y') + \xi(X')\eta(Y).$$

Proposition 6.11 (Separated coupled kernel). *For every background-mixed unit pair X, Y ,*

$$(91) \quad \mathfrak{m}_{B_{\xi, \eta}}(X, Y) = (\nabla_X \xi)(X)(\nabla_Y \eta)(Y).$$

Consequently $B_{\xi, \eta}$ lies in the mixed two-jet kernel if and only if at least one of ξ and η is a Killing one-form. If ξ is Killing and $\eta = du$, then

$$(92) \quad B_{\xi, du} = \mathcal{L}_{u\xi^\sharp} h,$$

so the exact part of the second one-form is gauge.

Suppose ξ is Killing and $h + sB_{\xi, \eta}$ is Riemannian. Put

$$(93) \quad \mathcal{C}_{XY} = \frac{1}{2}(\eta(Y)d\xi(X, \cdot), \xi(X)d\eta(Y, \cdot)).$$

Then every background-mixed plane satisfies the exact identity

$$(94) \quad \sec_{h+sB_{\xi,\eta}}(X \wedge Y) = \frac{s^2}{\Delta_s(X, Y)}(h + sB_{\xi,\eta})^{-1}(\mathcal{C}_{XY}, \mathcal{C}_{XY}) \geq 0.$$

If $M_1 = \mathbb{S}^2$, $\xi \neq 0$, and $\dim M_2 \geq 2$, every such ray nevertheless has a background-mixed zero-curvature plane.

Proof. Use product-normal extensions of X and Y . The two pure blocks of $B_{\xi,\eta}$ vanish, while differentiating its mixed block twice gives (91). A one-form ξ is Killing exactly when $(\nabla_X \xi)(X) = 0$ for every X . Since the two factors in (91) depend on independent variables, their product vanishes identically exactly when one of the one-forms is Killing. Formula (92) follows directly from the Killing equation.

Let $\mathcal{C}$ be the first variation of the Christoffel symbol of the first kind. The Killing equation gives

$$\mathcal{C}_{XX} = 0, \quad \mathcal{C}_{XY} = \frac{1}{2}(\eta(Y)d\xi(X, \cdot), \xi(X)d\eta(Y, \cdot)).$$

The second-derivative part of the lower-index mixed curvature is linear in s and vanishes by (91). The exact remaining term is therefore

$$s^2(h + sB_{\xi,\eta})^{-1}(\mathcal{C}_{XY}, \mathcal{C}_{XY}),$$

because the term pairing $\mathcal{C}_{XX}$ with $\mathcal{C}_{YY}$ is zero. Division by the squared area proves (94).

A nonzero Killing field on the round 2-sphere is rotational and has a zero x . At (x, y) choose a nonzero $Y \in \ker \eta_y$, which is possible when $\dim M_2 \geq 2$. Then (93) vanishes for every X , giving the final assertion. $\square$

The exact parts in (92) disappear in the quotient by diffeomorphisms. On a sphere the Hodge decomposition therefore leaves the co-closed part of the other one-form as the first genuinely coupled separated sector. The following finite sum is the most symmetric such tensor on $\mathbb{S}^2 \times \mathbb{S}^2$.

Identify the two ambient Euclidean spaces with $\mathbb{R}^3$. For $a \in \mathbb{R}^3$, let $K_a(x) = a \times x$ and $L_a(y) = a \times y$, and denote their metric-dual one-forms by κ_a and λ_a . For an orthonormal basis a_0, a_1, a_2 of $\mathbb{R}^3$, set

$$(95) \quad B_{\text{rot}} = \sum_{r=0}^2 B_{\kappa_{a_r}, \lambda_{a_r}}.$$

Equivalently, its only nonzero block is

$$(96) \quad B_{\text{rot}}((X, 0), (0, Y)) = \langle x \times X, y \times Y \rangle.$$

Theorem 6.12 (A nongauge kernel ray with an oblique obstruction). *The tensor B_{rot} is trace-free and divergence-free. It is nonzero and L^2 -orthogonal to all Lie derivatives and to all tensors pulled back from the individual factors. In particular, it represents a genuinely coupled, nongauge element of the mixed two-jet kernel.*

The metric

$$k_s = h + sB_{\text{rot}}$$

has the sharp Riemannian range

$$k_s > 0 \iff |s| < 1.$$

Throughout this interval every background-mixed sectional curvature is non-negative. At each diagonal point (x, x) every background-mixed sectional curvature is zero. Nevertheless, fix an orthonormal basis (e_0, e_1, e_2) of $\mathbb{R}^3$ and put $p = (e_0, e_0)$. For $i = 1, 2$, define

$$E_i^+ = \frac{(e_i, e_i)}{\sqrt{2(1+s)}}, \quad E_i^- = \frac{(e_i, -e_i)}{\sqrt{2(1-s)}}.$$

The k_s -orthonormal plane spanned by

$$(97) \quad U = \frac{E_1^+ - E_1^-}{\sqrt{2}}, \quad V = \frac{E_2^+ + E_2^-}{\sqrt{2}}$$

has

$$(98) \quad \sec_{k_s}(U \wedge V) = -\frac{3s^2}{2(1-s)^2(1+s)} < 0 \quad (0 < |s| < 1).$$

Thus this canonical transverse coupled kernel ray leaves the nonnegative-curvature cone immediately, despite improving no fixed mixed plane in the negative direction.

Proof. Each κ_{a_r} and λ_{a_r} is Killing, so Proposition 6.11 and linearity give $\mathfrak{m}_{B_{\text{rot}}} = 0$. The tensor is pure mixed, hence trace-free. Its divergence vanishes because every Killing one-form is co-closed. Integration by parts makes a divergence-free symmetric tensor L^2 -orthogonal to every Lie derivative; pointwise block orthogonality gives the assertion for product tensors.

The maps $X \mapsto x \times X$ and $Y \mapsto y \times Y$ are isometries on their tangent planes. Hence

$$|B_{\text{rot}}((X, 0), (0, Y))| \leq \|X\| \|Y\|.$$

It follows that $k_s > 0$ for $|s| < 1$. Equality is attained in this estimate when $x = y$ and $X = Y$, so a symmetric or antisymmetric vector becomes null at $s = -1$ or $s = 1$, respectively; outside the closed interval the metric is indefinite. This proves the sharp range.

The proof of (94) applies unchanged to a sum whose first one-forms are Killing, proving mixed nonnegativity. For a rotational Killing form,

$$d\kappa_a(X, Z) = 2 \langle a, X \times Z \rangle.$$

At a diagonal point the two components of the summed covector $\mathcal{C}_{XY}$ contain scalar products between a tangent vector of the form $x \times Y$ and a normal vector of the form $X \times Z$; they vanish. Hence all background-mixed curvatures there are zero.

Finally use orthographic coordinates

$$x = (\sqrt{1 - |u|^2}, u_1, u_2), \quad y = (\sqrt{1 - |v|^2}, v_1, v_2)$$

centered at p . The mixed block is

$$(B_{\text{rot}})_{i\alpha} = \langle x \times \partial_{u_i} x, y \times \partial_{v_\alpha} y \rangle.$$

Substitution into the lower-index curvature formula for the vectors in (97) gives unit squared area and (98). $\square$

Corollary 6.13 (The diagonal rotational repair fails quartically). *Let $f \in C^\infty(\mathbb{S}^2 \times \mathbb{S}^2)$, put $Z = \nabla f$, and let Ψ_t be the flow of $Z/2$. Then*

$$(99) \quad g_t = \Psi_t^*(h + t^2 B_{\text{rot}}) = h + t \text{Hess } f + t^2 \left(\frac{1}{8} \mathcal{L}_Z^2 h + B_{\text{rot}} \right) + O(t^3).$$

This path satisfies every quadratic mixed repair identity in Theorem 6.2. Nevertheless, for every $0 < |t| < 1$ it contains a plane of curvature

$$(100) \quad -\frac{3t^4}{2(1-t^2)^2(1+t^2)} < 0.$$

Proof. The expansion follows from pullback by the flow. Its normalized second-order tensor is B_{rot} , which lies in the mixed two-jet kernel. Pullback preserves sectional curvature, so substitute $s = t^2$ in (98). $\square$

7. CONSEQUENCES, CONTEXT, AND SCOPE

7.1. The four-dimensional Hopf case. On $\mathbb{S}^2 \times \mathbb{S}^2$, every bilinear product of degree-one spherical harmonics has the form s_A . Orthogonal changes of coordinates in the two factors reduce A to singular-value form, so Theorem 1.1 is a classification-free result for the whole nine-dimensional space of such potentials. On the open dense stratum $\sigma_1 > \sigma_2$, Theorem 3.5 detects the negative curvature on an explicit background-mixed plane. On the exceptional stratum $\sigma_1 = \sigma_2$, the flat-torus theorem supplies three embedded zero-curvature loci, while Proposition 3.9 detects strict negative curvature on an explicit oblique plane. At the isotropic endpoint $\sigma_1 = \sigma_2 = \sigma_3$, Theorem 3.7 goes further and proves that every background-mixed curvature is nonnegative on the full Riemannian interval. Thus the isotropic endpoint makes the distinction between background-mixed and oblique planes unavoidable. Specializing Corollary 3.11 shows that every nontrivial ray in the whole nine-dimensional family leaves the nonnegative-curvature cone immediately. The top-critical oblique mechanism works throughout rank at least two, the zero-critical oblique mechanism handles rank one, and the simple-top stratum additionally admits the background-mixed spectral-gap mechanism. Corollary 6.3 strengthens the local conclusion: after the infinitesimal diffeomorphism has been removed, every nonzero ray already has a negative quadratic mixed-curvature coefficient. Thus the quartic spectral-gap plane is a special background-mixed phenomenon, not the first obstruction among all planes.

This conclusion is independent of the Hsiang–Kleiner theorem. In the generic full-rank case the inherited background symmetry is finite, so one cannot obtain the conclusion merely by pointing to an inherited circle action;

Theorem 3.5 in fact gives strict negative curvature there. Conversely, Corollary 5.2 records that the repeated-top case does retain an inherited circle action, but the explicit oblique calculation already proves negative curvature there. We emphasize again that this does not prove the full isometry group is finite in the generic case; it shows only that the direct curvature and flat-torus arguments do not need continuous inherited symmetry there.

7.2. Relation to deformations of product metrics. Bourguignon's work placed deformations of product metrics into the broader study of Hopf's conjecture [5, 4]. The family considered here is a particular exact Hessian deformation through the round product. Its special feature is not a general stability theorem for product metrics, but an algebraic compatibility between singular two-planes and fixed-point geometry. That compatibility persists for the entire interval (8), including parameters far from the infinitesimal regime.

The mixed-curvature calculation adds a complementary point. On each fixed background-mixed plane its second-order term is nonnegative, which is compatible with the classical infinitesimal analysis of product variations [4]. Theorem 3.5 shows that when this coefficient vanishes, the first negative term on a distinguished background-mixed plane can be quartic. Theorem 6.1 explains why this does not make the full metric deformation second-order nonnegative: after gauge normalization, a moving oblique plane becomes a fixed mixed plane, and Corollary 6.3 gives its negative quadratic coefficient. A nonlinear continuation can therefore survive the two-jet test only if its correction tensor B cancels the quadratic mixed-curvature functional everywhere. The integrated identity (51) forces equality, and the singular planes give the explicit costs (57) and (58). The canonical gauge correction (55) shows that this system is consistent, but by itself produces no new geometry. Theorem 6.4 identifies the entire quadratic kernel modulo gauge and product tensors; on $\mathbb{S}^2 \times \mathbb{S}^2$ it leaves only the bidegrees $(1, \ell)$ and $(\ell, 1)$. Theorem 6.9 tests every nonzero element of that quotient at once: a moving plane has negative curvature at order s^2 , even for cross-degree superpositions, simultaneous two-tower mixtures, and arbitrary $(1, 1)$ components. The corresponding canonical corrected Hessian path fails at order t^4 . Proposition 6.11 gives the exact nonlinear mixed-curvature formula for its separated elements. The rotational tensor in Theorem 6.12 is transverse to the diffeomorphism orbit and makes every fixed background-mixed curvature nonnegative. Its explicit oblique curvature is nevertheless negative at order s^2 , or order t^4 in the corrected Hessian path of Corollary 6.13. Thus passing the complete quadratic mixed test and the ensuing fixed-mixed quartic test is still not enough to control nearby oblique planes.

Symmetry-based obstructions remain among the strongest general results related to Hopf's question. Hsiang and Kleiner proved that a positively curved oriented four-manifold with an isometric circle action has $b_2 \leq 1$ [7]; Amann and Kennard established higher-dimensional product obstructions under

quantitative torus symmetry assumptions [2]. Other authors have treated special metric forms, including conformal and doubly warped products [8]. Our result is complementary: the metrics generally have mixed Hessian blocks and need not be conformal or warped, while their obstruction is an explicit flat submanifold rather than a conclusion drawn from an effective torus action.

7.3. Higher-dimensional content and limitations. The construction and the curvature isolation theorem are dimension-independent. For $q = \min\{m+1, n+1\}$ it yields at least $\binom{q}{2}$ tori for each singular frame, so the number of certified zero-curvature loci grows quadratically with the smaller ambient dimension. This gives a family-level obstruction on every product of positive-dimensional spheres. When both spheres have dimension at least two, the singular-subsphere reduction promotes the four-dimensional top-critical calculation to every ambient dimension. Combined with the rank-one zero-critical calculation, it shows that every nontrivial bilinear Hessian ray leaves the nonnegative-curvature cone immediately, with the failure always detected inside the top-three singular $\mathbb{S}^2 \times \mathbb{S}^2$. Corollary 3.12 upgrades this qualitative statement to an explicit dimension-free negative bound on every normalized annulus away from the product metric and the Riemannian boundary.

The result should not be read as a curvature classification of $g_{\gamma,A}$. We give an exact formula for every background-mixed plane and determine a distinguished collection of zero sectional curvatures. Proposition 3.9 determines one distinguished oblique plane for every matrix of rank at least two, which is enough, together with the rank-one zero-critical plane, for complete ray isolation in all dimensions at least two. We do not classify all planes whose spanning vectors both have components in both factors. Nor do Hessian deformations $h + \gamma \text{Hess } s_A$ exhaust all metrics on a product of spheres. The two-jet theorem constrains arbitrary nonlinear paths sharing the same Hessian first variation, and the integrated theorem shows that a nonnegative path must cancel every quadratic mixed-curvature coefficient. This cancellation is feasible: the gauge correction does it identically, and product variations lie in the kernel as well. Theorem 6.4 classifies the full quotient and shows that one harmonic leg must be Killing. Proposition 6.11 gives the exact background-mixed curvature of separated representatives, and Theorem 6.9 obstructs every nonzero coupled kernel ray, including arbitrary mixtures of all allowed harmonic sectors. Theorem 6.12 settles the canonical $(1,1)$ sum by an exact formula. What remains open is not a sector of the quadratic kernel, but the higher-jet problem: arbitrary third- and higher-order corrections can enter before or at the quartic obstruction of a canonical corrected path. A broader obstruction would require compatibility conditions on those higher jets, a substantially larger class of first variations, or a full curvature analysis. The present kernel-ray results are sharp in their stated class.

7.4. Verification. The arguments combine the Hessian identity (2), the two-frame singular-value principle, the exact mixed-curvature identity, and the fixed-point lemma for isometries. As an independent check against coordinate or sign errors, the source distribution includes the script `verify_hessian_tori.py`. It constructs the induced torus metric from (40), computes its Levi-Civita connection and Riemann tensor directly, and verifies symbolically that its Gaussian curvature is zero. It also checks the separated normal form and the sharp endpoint degeneracies. The companion script `verify_mixed_curvature_counterexample.py` differentiates the full orthographic-coordinate metric symbolically and checks the mixed-curvature identity, the spectral-gap formula, the simple zero-threshold specialization, the plane-wise nonzero threshold, and the isotropic quaternion factorization, the uniform top-critical oblique-plane curvature, its two-jet coefficient, and the rank-one zero-critical specialization using exact arithmetic. The separate script `verify_killing_kernel.py` checks the diagonal rotational kernel metric, its sharp Riemannian boundary, its flat diagonal mixed plane, and its exact negative oblique curvature. The script `verify_coupled_kernel_classification.py` checks the factorization through the two Killing operators, the tensor-product nullity formula, and the gauge/co-closed normal form in an exact finite-mode model. Finally, `verify_coupled_kernel_obstruction.py` differentiates the exact orthographic-coordinate metric, recovers the general two-parameter moving-plane coefficient, tests a nonseparated degree-two vector harmonic and an exact cross-degree frame average, and verifies the signed-SVD obstruction for an arbitrary $(1, 1)$ block.

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